MATHEMATICS GRADE 10: CONGRUENCE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.4 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 1.0: Numbers
Sub-Strand	Sub-Strand 1.4: Congruence
Total Duration	8 lessons (approximately 16 periods × 40 minutes = 640 minutes)
Sub-Strand Content	– Lines of symmetry: identifying and drawing lines of symmetry in 2D shapes and real-world objects – Reflection on the Cartesian plane: reflecting points and objects along the x-axis and y-axis – Reflection along lines parallel to the x-axis and y-axis (e.g., $y = k$ and $x = k$) – Reflection along the lines $y = x$ and $y = -x$ – Finding the equation of a mirror line (line of reflection) given an object and its image – Congruence: defining and identifying congruent figures; congruence conditions for triangles (SSS, SAS, ASA, AAS, RHS) – Applications of reflection and congruence in real-life situations (art, architecture, fabric design, nature)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify and draw lines of symmetry in two-dimensional shapes and real-world objects found in Kenyan contexts; b) reflect points and plane figures on the Cartesian plane along the x-axis, y-axis, lines parallel to the axes, and the lines $y = x$ and $y = -x$; c) determine the equation of a mirror line given a figure and its reflected image on the Cartesian plane; d) define congruence and apply congruence conditions (SSS, SAS, ASA, AAS, RHS) to identify and prove that two triangles are congruent; e) apply knowledge of reflection and congruence to solve problems in real-life situations such as design, architecture, and nature.
Core Competencies	– Communication and Collaboration: Learners discuss and justify their reasoning about lines of symmetry, reflection transformations, and congruence conditions with peers during group activities and class presentations. – Critical Thinking and Problem Solving: Learners analyse Cartesian graphs to determine mirror lines, apply congruence conditions to unknown figures, and solve multi-step real-life problems involving reflections. – Creativity and Imagination: Learners design symmetrical patterns inspired by Kenyan kitenge/ankara fabric, Maasai beadwork, and architectural motifs, applying reflection and congruence principles. – Digital Literacy: Learners engage with videos and simulations on the ARES platform to observe reflections dynamically, building fluency with digital mathematical tools. – Self-Efficacy: Learners build confidence by progressing from familiar symmetry (lines of symmetry) to formal congruence proofs, celebrating incremental mastery at each lesson phase.
Core Values	– Respect: Learners appreciate the mathematical beauty found in Kenyan cultural designs (Maasai beadwork, kikoi patterns, mosque architecture in Mombasa) that rely on symmetry and congruence, fostering respect for diverse cultural heritage.

	– Responsibility: Learners take ownership of constructing accurate reflections and congruence arguments, understanding that precision in mathematics has real consequences in engineering and design professions. – Unity: Collaborative group tasks (mirror-line discovery, congruence matching activities) build a sense of shared inquiry and mutual support among learners of varied abilities. – Integrity: Learners present honest mathematical reasoning and acknowledge errors during model revision, modelling academic integrity in problem-solving.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate questions on the Driving Question Board about why reflected images appear identical yet reversed, and what conditions guarantee two shapes are truly congruent. – Developing and Using Models: Learners iteratively build and revise Cartesian plane models showing object-image relationships under successive reflections, updating them as new mirror lines are introduced each lesson. – Planning and Carrying Out Investigations: Learners carry out hands-on mirror-line discovery activities using folded paper, reflective surfaces, and graph-paper constructions to gather evidence about invariant properties under reflection. – Analysing and Interpreting Data: Learners compare coordinates of objects and images across multiple reflection scenarios, identifying numerical patterns (e.g., $(x, y) \rightarrow (-x, y)$ for y-axis reflection) to generalise transformation rules. – Constructing Explanations and Designing Solutions: Learners construct formal congruence arguments using identified conditions and apply them to design symmetrical structures or solve missing-measurement problems. – Engaging in Argument from Evidence: Learners defend their choice of congruence condition (SSS, SAS, ASA, AAS, or RHS) using measured side and angle evidence, critiquing peers' justifications respectfully.
Pertinent & Contemporary Issues (PCIs)	– Citizenship: Learners explore how symmetry and congruence appear in Kenyan national symbols (the Kenyan flag's symmetry, coat of arms design), connecting mathematics to civic identity and pride. – Social Cohesion and Peacebuilding: Collaborative tasks that require consensus on correct mirror lines and congruence proofs model peaceful negotiation and shared decision-making. – Environmental Education: Learners examine symmetry in Kenyan flora and fauna (butterfly wings, Nile tilapia fish, acacia tree silhouettes, Lake Victoria shoreline reflections) to appreciate mathematical order in nature. – Financial Literacy: Learners discuss how architects, textile designers, and tilers in Kenya use congruence to minimise material waste and maximise efficiency, linking mathematics to economic decision-making.
Career Connections	– Architecture and Construction: Architects designing buildings in Nairobi's CBD or the Konza Technopolis use symmetry and congruent structural elements to ensure aesthetic balance and structural integrity. – Fashion and Textile Design: Kenyan kitenge and ankara fabric designers apply reflective symmetry and congruent pattern units to produce repeating decorative motifs. – Surveying and Land Mapping: Land surveyors use congruence conditions to calculate inaccessible distances and verify plot boundaries in counties across Kenya. – Graphic Design and Animation: Digital artists and animators at Kenyan studios use reflections and congruent shapes as foundational tools for creating symmetrical logos, characters, and visual effects. – Engineering and Manufacturing: Engineers at facilities such as Kenya's Standard Gauge Railway workshops apply congruence to ensure interchangeable, identically manufactured parts fit precisely.
Focus for Lessons	This unit anchors the study of congruence within the broader experience of reflection symmetry — a property learners can see in everyday Kenyan life, from the mirrored surface of Lake Victoria at dawn to the symmetrical patterns of Maasai beadwork. Learners begin by exploring lines of symmetry intuitively, then systematically investigate reflections on the Cartesian plane across progressively more complex mirror lines, building towards a rigorous understanding of congruence conditions for triangles. Throughout the unit, an inquiry-driven storyline connects each new reflection rule back to the anchoring phenomenon, ensuring learners see congruence not as an isolated geometric fact but as the mathematical explanation for why reflected images are

	always perfect, size-preserving copies of their originals.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy — and how can we use that same idea to prove that two shapes are truly identical?" KICD KEY INQUIRY QUESTIONS: 1. How do you determine the line of symmetry of a figure? 2. How do you apply reflection to real-life situations?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Perfect Mirror Image at Lake Victoria's Shore" At dawn on the shore of Lake Victoria near Kisumu, fishermen and their boats cast perfect mirror images on the still water below. The reflected image appears identical in shape and size to the real boat above the waterline — yet it is flipped. Students observe (via a projected photograph and a short teacher-led demonstration using a flat mirror and a cut-out boat shape) that no matter how complex the shape of the boat, its reflection is always the same size and always appears at an equal distance on the other side of the waterline. WHY IT IS PUZZLING: Students are puzzled by several things at once: (1) Why does the reflection always look exactly the same size — neither bigger nor smaller — than the original? (2) What is the "rule" the water surface is following to produce the image? (3) If you moved the boat further from the water surface, would the reflection move the same amount? (4) Could you predict exactly where every point of the reflected image would land? (5) What does it mean, mathematically, for two shapes to be "exactly the same"? These questions sustain inquiry across all 8 lessons as students move from intuitive symmetry to formal congruence proofs.
Storyline Thread	Lesson 1 — "Finding Balance: Lines of Symmetry" | Anchoring phenomenon introduced (Lake Victoria mirror image). Students observe the photograph, record initial predictions about why the reflection is the same size, and post opening questions on the Driving Question Board. Hands-on activity: students fold cut-out shapes (fish, boat, acacia leaf) to find lines of symmetry. Video: lines of symmetry in Kenyan objects. Students update their DQB with new questions about what happens when you "flip" a shape mathematically. Lesson 2 — "Flipping Across the Axes: Reflection on the Cartesian Plane (x- and y-axes)" | Students watch the video on reflecting points on the Cartesian plane (x-axis) and the video of reflecting objects. They plot boat-shape vertices on a grid, reflect across the x-axis and y-axis, and record coordinate pairs in a table to discover the transformation rules $(x, y) \rightarrow (x, -y)$ and $(x, y) \rightarrow (-x, y)$. Connection to phenomenon: the waterline IS the x-axis — the image is always the same distance below as the object is above. DQB updated: "What if the mirror line is NOT the x- or y-axis?" Lesson 3 — "Shifting the Mirror: Reflection Along Lines Parallel to the Axes" | Students watch the video on reflection along $y = -4$ and similar lines. Using the matatu-map supporting phenomenon, students reflect route landmarks across $y = 2$, $y = -4$, $x = 3$, and $x = -1$, generalising the rules for parallel mirror lines. Sensemaking discussion: how is this different from axis reflection? Model update: students add parallel mirror lines to their cumulative Cartesian models. Lesson 4 — "Diagonal Mirrors: Reflection Along $y = x$ and $y = -x$" | Inspired by the Mombasa tile supporting phenomenon, students reflect points and polygons across $y = x$ and $y = -x$, discovering the rules $(x, y) \rightarrow (y, x)$ and $(x, y) \rightarrow (-y, -x)$. Predict-Observe-Explain cycle run in full. Students notice that diagonal reflections feel like "rotations" — a productive misconception to address.

	DQB: "How would we find the mirror line if we were only given the object and its image?" Lesson 5 — "Reverse Engineering the Mirror: Finding the Equation of the Mirror Line" | Students watch the video on finding mirror lines and complete the hands-on activity (finding mirror lines). Given object-image pairs plotted on a grid, students construct perpendicular bisectors of corresponding point pairs and determine the equation of the mirror line. Connection to phenomenon: "If you only saw the boat and its reflection — but the water surface was hidden — could you figure out where the water surface was?" DQB updated with equations discovered. Lesson 6 — "Same Shape, Same Size: Introducing Congruence" | Students watch the congruence video. Using the roof-truss supporting phenomenon, students measure triangles cut from card and sort them into "definitely the same" and "possibly the same" groups, motivating the need for formal congruence conditions. SSS, SAS, ASA, AAS, and RHS conditions introduced through guided investigation. Students link back to phenomenon: reflected images are ALWAYS congruent to their originals — and now they can prove it. Lesson 7 — "Proving It: Applying Congruence Conditions" | Students solve structured problems requiring them to identify which congruence condition applies and write formal congruence statements. Peer argumentation task: groups challenge each other's proofs. Connection to phenomenon: the mathematical reason the Lake Victoria boat reflection is the same size is formalised — reflection is a rigid transformation that preserves all side lengths and angles, guaranteeing congruence. Lesson 8 — "Mathematics in the Real World: Applications of Reflection and Congruence" | Students watch the applications video and work on real-life problems: designing a symmetrical Maasai bead pattern, calculating inaccessible distances using congruent triangles, and verifying a building plan. Final DQB review: students revisit every original question and confirm which have been answered. Cumulative model presented and explained. Exit task: students write a full explanation of the anchoring phenomenon using the mathematical language of reflection and congruence acquired across the unit.
Supporting Phenomena	– Maasai Beadwork Symmetry (Lesson 1 — Lines of Symmetry): Students examine high-resolution photographs of Maasai beaded jewellery and identify multiple lines of symmetry within the repeating pattern units, asking why certain designs look "balanced" from every angle. – Reflecting a Matatu Route Map on a Grid (Lessons 2–4 — Cartesian Reflections): A simplified Nairobi matatu route map is plotted on a Cartesian grid; students observe how reflecting the map across the x-axis, y-axis, and parallel lines produces a "mirror-city" and investigate which landmarks land where. – Mosque Geometric Tiling — Mombasa Old Town (Lesson 5 — $y = x$ and $y = -x$ Reflections): Students observe photographs of geometric tile patterns from the Old Town Mombasa mosques, noting that diagonal mirror lines produce rotated-looking copies, prompting inquiry into $y = x$ and $y = -x$ reflections. – Identical Roof Trusses on a Kenyan School Building (Lessons 6–7 — Congruence): A photograph of a newly constructed school in the Rift Valley shows a row of identical triangular roof trusses. Students ask: how did the builders ensure every truss is exactly the same without measuring every single dimension each time? This motivates the congruence conditions.

LESSON 1 (40 minutes): Finding Balance: Lines of Symmetry	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit by anchoring student curiosity in the Lake Victoria mirror-image phenomenon and establishing the foundational idea that a line of symmetry divides a figure into two mirror-image halves that are identical in shape and size.
Knowledge	Students will know that a line of symmetry divides a plane figure into two congruent halves that are mirror images of each other, and that a figure may have one, more than one, or no lines of symmetry.
Skills	Students will be able to identify and draw lines of symmetry on cut-out shapes by folding, explain why the two halves match exactly, and record initial questions about reflection and congruence on the Driving Question Board.
Attitudes	Students will appreciate the mathematical order found in everyday Kenyan environments and develop curiosity and open-mindedness as they encounter puzzling aspects of the phenomenon.
Key Inquiry Question	How do you determine the line of symmetry of a figure?
Purpose in Storyline	This is the opening lesson of the 8-lesson arc. It introduces the anchoring phenomenon (the Lake Victoria mirror image), surfaces prior knowledge and misconceptions, and plants the questions that will drive inquiry through all subsequent lessons. The hands-on folding activity gives students a physical, embodied entry point into the abstract idea of reflection before any formal definitions are introduced.
Safety Notes	Scissors are used during the cut-out activity — remind students to carry scissors with points facing downward and to cut away from their bodies. Ensure the projected image display does not use direct sunlight or laser sources that could affect vision.

B. LESSON OVERVIEW
Students are shown a projected photograph of a fishing boat at dawn on Lake Victoria near Kisumu whose mirror image appears perfectly on the still water surface. After a brief teacher-led demonstration with a flat mirror and a card cut-out of a boat, students record their initial predictions about why the reflection is always the same size. They then conduct a hands-on folding activity using cut-out shapes — a fish, a boat silhouette, and an acacia leaf — to discover lines of symmetry. A short video segment highlights lines of symmetry in familiar Kenyan objects (Maasai shields, beaded jewellery, the Kenyan flag). The lesson closes with the creation of the class Driving Question Board (DQB), where students post opening questions about reflection and congruence that will be investigated across the unit.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic] Source: CK-12	Students enter to find a high-quality projected photograph of a fishing boat on Lake Victoria at dawn, its mirror image perfectly visible on the still water. Students sit in groups of four. Each student receives a sticky note and is	Teacher points to the photograph and says: 'Before we say anything mathematical today, I want you to just look at this image. This was taken near Kisumu, on the shore of Lake Victoria, early in the morning when the water is	Individual think-write before group share (Think-Write-Pair-Share) ensures every student commits to a prediction before social influence. Teacher publicly records all predictions to honour student thinking and create an	Teacher scans sticky notes as groups form their group prediction. Look for: (1) students who mention equal distance from the water surface — these show early intuitive understanding of reflection distance; (2) students who say the

🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions: Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	asked: 'Look at this image carefully. Why do you think the reflection of the boat looks exactly the same size as the real boat — neither bigger nor smaller? Write your best prediction.' Students write individually for 90 seconds, then share within their group and agree on one group prediction to report.	perfectly still. Tell me — what do you notice? What do you wonder?' Allow 20 seconds of silent looking (WAIT TIME 1). Then: 'Now, on your sticky note, write your best scientific or mathematical prediction: why is the reflection exactly the same size as the boat?' After individual writing, ask one reporter per group to share. Teacher listens without evaluating, recording key prediction phrases on the board under the heading PREDICTIONS — DO NOT ERASE. Teacher then says: 'These are your ideas right now. By the end of this unit you will be able to prove whether each prediction is correct using formal mathematics.'	accountability anchor revisited in Lesson 8.	water acts like a mirror — connect this to the formal mirror-line concept in the Explain Phase; (3) students who predict size changes based on depth — note as a productive misconception to address in Lesson 2.
Observe Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions: Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Phase 1 (Teacher demonstration — 5 minutes): Teacher holds a card cut-out of a boat shape against a flat mirror placed horizontally on the document camera or desk. Students observe how the image appears on the other side of the mirror surface, at what appears to be the same distance. Teacher moves the boat closer and further from the mirror and asks: 'What do you observe about the image as I move the boat?' Students respond chorally or with hand signals (thumbs up: image gets closer to mirror too; thumbs down: image stays in place). Phase 2 (Hands-on folding — 12 minutes): Each group receives three pre-cut card shapes: a Nile tilapia fish outline, a simplified dhow boat silhouette, and an acacia leaf outline. Students fold each shape trying to make the two halves match exactly. For each successful fold they draw the fold line with a pencil	During Phase 1 demonstration, teacher says: 'Watch carefully — I am not going to tell you what is happening yet. Your job is to observe and be a detective.' After moving the boat, pause (WAIT TIME 2 — 10 seconds) before taking responses. During Phase 2, circulate and use targeted questions: 'How do you know the two halves match exactly?' and 'Is there more than one way to fold this shape so the halves match?' If a group finds only one line on the acacia leaf, prompt: 'Try folding it the other way — does it still match?' After the video, teacher asks: 'Which object surprised you most? Why?' (WAIT TIME 3 — 15 seconds of silent thinking before hands are raised.)	Physical folding provides kinesthetic and visual proof of symmetry — students feel the halves aligning. The three shapes are carefully chosen for increasing complexity: the fish has one line, the boat has one line, and the acacia leaf can stimulate debate about whether it has exactly one or approximately one (introducing the idea that mathematical symmetry is exact, unlike natural objects). The video grounds abstract mathematics in Kenyan cultural and natural contexts.	As students fold, teacher checks: Can students articulate WHY a fold line is a line of symmetry (the two halves are mirror images)? Can they distinguish a line of symmetry from any random fold? Collect the folding-record tables at end of the observation phase. Use them to identify: (1) students who could not find any line of symmetry on the fish (may need one-to-one support in Lesson 2); (2) students who claim the acacia leaf has multiple symmetry lines — use as a discussion point about exact versus approximate symmetry.

	and label it 'Line of Symmetry.' Students record results in a simple table: Shape, Number of Lines of Symmetry Found, Sketch of the Line(s). Phase 3 (Video — 3 minutes): Teacher plays a short pre-prepared video showing lines of symmetry in Kenyan objects: a Maasai beaded collar, a kikoi fabric pattern, the outline of Mount Kenya, and the Kenyan coat of arms shield.			
Explain Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions: Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students return to the projected Lake Victoria photograph. The teacher introduces the term LINE OF SYMMETRY formally, writing the definition on the board: 'A line of symmetry is a line that divides a figure into two congruent halves that are mirror images of each other.' Teacher then asks students to identify the line of symmetry in the photograph (the waterline). Teacher connects the folding activity to the photograph: 'When you folded your fish and the two halves matched exactly — that fold line is a line of symmetry. In the Lake Victoria image, the waterline IS the fold line. The real boat and its reflection are the two halves.' Students are then asked: 'So what word might we use to describe two shapes that are exactly the same in shape and size?' Teacher introduces the word CONGRUENT (written on the board) without formal definition yet — just as a word for students to carry forward. Students write a 2-sentence explanation in their notebooks: (1) what a line of symmetry is in their own words, and (2) one question the explanation raises in their mind.	Teacher says: 'Let us connect what you observed during the folding back to the photograph. Look at the waterline in the image.' Point explicitly to the waterline. 'In mathematics, what role is this waterline playing? Talk to your partner for 30 seconds.' (WAIT TIME 4 — 30 seconds partner talk.) After responses, teacher says: 'Exactly — it is acting as the line of symmetry. And here is a new mathematical word for you: when two shapes are exactly the same in shape and size, we call them CONGRUENT. Write that word down. We will be building a full definition over the next several lessons.' Teacher explicitly says: 'I am not giving you the full definition today because I want you to build it yourself through investigation. What we are doing today is just opening the door.' Teacher avoids confirming or rejecting student predictions from the Predict Phase — instead says: 'Keep those predictions in mind. We will return to them.'	Bridging talk (explicit verbal connection between the folding activity and the photograph) is used to help students transfer their kinesthetic folding experience to the abstract visual context. Introducing CONGRUENT as a vocabulary item without a full formal definition is intentional — it primes students for sensemaking in Lessons 6 and 7 without closing down inquiry prematurely.	Collect the 2-sentence notebook entries. Look for: (1) students who can express that a line of symmetry creates two identical halves — indicates readiness for Lesson 2 reflection rules; (2) students who conflate symmetry with equal area (the two halves have equal area but that is not sufficient — they must also be mirror images) — address this explicitly in the next lesson; (3) quality of the question students write — rich questions become candidates for the DQB.
Driving	Each student receives two sticky	Teacher says: 'A mathematician is	The DQB is a visible, persistent	The sticky notes are themselves

Question Board (DQB) Creation 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions: Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	notes of different colours. On the FIRST sticky note (colour A), they write one question about the Lake Victoria phenomenon that they cannot yet answer. On the SECOND sticky note (colour B), they write one question about lines of symmetry or reflection that arose from the folding activity or the explanation. Students bring their sticky notes to a large sheet of chart paper taped to the wall, labelled DRIVING QUESTION BOARD — OUR QUESTIONS ABOUT REFLECTION AND CONGRUENCE. Teacher reads several aloud and groups similar questions together into clusters, labelling emerging clusters with headings such as: WHY SAME SIZE?, WHAT IS THE RULE?, WHAT IF THE LINE MOVES?, HOW DO WE PROVE IT? The driving question of the unit is displayed prominently above the DQB: 'How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy — and how can we use that same idea to prove that two shapes are truly identical?'	someone who is puzzled by the world and refuses to leave a question unanswered. Right now I want you to be puzzled. Write your most burning question on your first sticky note — a question about the boat and its reflection that you cannot answer yet.' Give 60 seconds of silent writing (WAIT TIME 5). Then: 'On your second sticky note, write a question that the folding activity or the explanation raised for you.' After students post their notes, teacher reads five or six aloud with genuine curiosity in their voice: 'Oh — this one asks: if you moved the boat higher above the water, would the reflection go deeper below? That is an extraordinary question. Let us put it here under WHY SAME SIZE.' Teacher says: 'We will return to this board at the end of every lesson. By Lesson 8, every question on this board should have an answer — and YOU will have put it there.'	record of the class community of inquiry. Using two different sticky-note colours ensures students generate questions at two levels: phenomenon-level (the boat image) and concept-level (symmetry and reflection). Clustering questions on the board in front of students makes the structure of the upcoming inquiry arc visible and gives students a sense of ownership over the direction of learning.	formative data. Teacher photographs the completed DQB at the end of the lesson. Review the photographs to: (1) identify which conceptual gaps are most widespread (clusters with many notes); (2) identify students whose questions reveal strong prior knowledge (use as peer resources in Lesson 2); (3) identify students whose questions are very surface-level or off-topic (may need additional engagement or language support). These findings should shape the opening of Lesson 2.
Model Building Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions:	Each student opens their Mathematics notebook to a clean double page headed: MY GROWING MODEL OF REFLECTION AND CONGRUENCE. Teacher models on the board what this page should contain at the end of Lesson 1. Students draw or paste the following onto their model page: (1) A simple sketch of the Lake Victoria boat photograph with the waterline labelled LINE OF SYMMETRY; (2) Their best sketch of one of the folded shapes with	Teacher says: 'Scientists and mathematicians keep working models — not final answers, but a record of their best thinking at this moment. Your model page is not meant to be complete today. It is meant to be honest.' Display a teacher-prepared example model page on the document camera. Point to the incomplete definition box for CONGRUENT and say: 'See how I have left this box open? That is deliberate. We do not have the full answer yet — and that is not a problem. That is the whole	The growing model is a metacognitive scaffold — it makes student thinking visible to themselves over time. Beginning the model in Lesson 1 with intentionally incomplete sections motivates the remaining lessons: students have a personal stake in filling in the blanks. The lake Victoria sketch as the anchor image on every model page maintains the phenomenon connection across all 8 lessons.	Quick scan of model pages before students close notebooks. Teacher looks for: (1) Correct placement and labelling of the line of symmetry on the boat sketch — indicates the key concept has landed; (2) Whether students have carried their genuine DQB question onto the model page — indicates personal investment in the inquiry; (3) Neatness and organisation are NOT assessed — the model is a thinking tool, not a presentation task. Verbally acknowledge two or three

Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	the fold line drawn and labelled; (3) The word CONGRUENT in a box with the note: two shapes that are exactly the same — definition to be completed; (4) The question they most want to answer, copied from their DQB sticky note. Teacher tells students: 'This model page will grow across all 8 lessons. By Lesson 8 it will contain the full mathematical explanation of why the Lake Victoria image is always a perfect, same-sized copy.'	point of the next seven lessons.' Circulate as students work, giving specific encouragement: 'That sketch is very clear — well done.' Avoid correcting the model at this stage unless a student has written something factually harmful to future learning (e.g., writing that reflections always make images smaller — gently redirect: 'Look back at the demonstration — did the image change size?'). Close by saying: 'Put your model pages somewhere safe. You will need them next lesson when we take the boat to the Cartesian plane.'		exemplary model pages by describing what makes them strong as mathematical thinking records (not artistic quality).

D. TEACHER REFLECTION

After the lesson, consider: Did students generate genuine questions for the DQB, or did they write what they thought I wanted to hear? Were the three folding shapes (fish, boat, acacia leaf) appropriately challenging — or did groups finish too quickly, suggesting a need for a more complex fourth shape in future iterations? Did the introduction of the word CONGRUENT feel premature, or did students receive it with curiosity? Were there students who were reluctant to engage with the photograph — might they need a more personally familiar phenomenon context? Review the DQB photograph and the notebook model pages before Lesson 2 to identify the two or three questions that most naturally lead into reflection on the Cartesian plane, so that the opening of Lesson 2 explicitly honours those questions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that the Lake Victoria boat reflection appears exactly the same size as the real boat, that the waterline divides the image into two matching halves, and that folded cut-out shapes (fish, boat, acacia leaf) have fold lines along which the two halves align exactly.
What did I learn?	Students learned that a line of symmetry is a line that divides a figure into two congruent halves that are mirror images of each other, that the waterline in the Lake Victoria photograph acts as a line of symmetry, and that the word CONGRUENT describes two shapes that are exactly the same in shape and size.
How does this explain the phenomenon?	Students began to explain that the reason the reflected boat image is the same size as the original is connected to the way the waterline acts as a line of symmetry — but the full mathematical explanation (reflection as a rigid transformation preserving all lengths and angles) remains open for investigation across the remaining lessons.

LESSON 2 (60 minutes): Flipping Across the Axes: Reflection on the Cartesian Plane (x- and y-axes)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will discover and apply the coordinate rules for reflecting points and figures across the x-axis and y-axis on the Cartesian plane, connecting the transformation rules to the Lake Victoria mirror-image phenomenon.
Knowledge	Students will know that: (1) reflecting a point across the x-axis maps (x, y) → (x, −y); (2) reflecting a point across the y-axis maps (x, y) → (−x, y); (3) each point in the original figure is the same perpendicular distance from the line of reflection as its corresponding point in the image; (4) the image produced by a reflection is congruent to the original figure.
Skills	Students will be able to: (1) plot vertices of a polygon on the Cartesian plane; (2) reflect a set of points and a composite figure across the x-axis and the y-axis; (3) record and compare coordinate pairs in a table to identify transformation rules; (4) measure perpendicular distances from a point to the line of reflection and verify equal distances on both sides.
Attitudes	Students will appreciate the precision and predictability of mathematical rules by recognising that the 'behaviour' of the Lake Victoria reflection is governed by an exact coordinate rule — and that mathematics can describe natural phenomena accurately.
Key Inquiry Question	How do you apply reflection to real-life situations?
Purpose in Storyline	In Lesson 1 students identified lines of symmetry intuitively using the Lake Victoria photograph and a flat mirror. In Lesson 2 students move from intuition to precision: they place the boat shape on a Cartesian grid (the waterline becomes the x-axis) and discover algebraically what the water surface is 'doing' to every point. This sets up Lesson 3, where the mirror line will no longer be one of the axes.
Safety Notes	No physical hazards. Ensure printed grid worksheets are distributed before the lesson to avoid time lost. If using scissors to cut boat templates, remind students to handle them carefully and pass them point-down.

B. LESSON OVERVIEW

Lesson 2 sits at the heart of the evidence-gathering sequence. Students begin by revisiting the Lake Victoria phenomenon — the waterline acting as a perfect mirror — and are challenged to place that real-world situation onto a Cartesian grid. Working in pairs, they plot the five vertices of a 'boat' polygon above the x-axis, reflect the shape across the x-axis (the waterline), and record every original and image coordinate in a two-column table. Through guided questioning they articulate the rule (x, y) → (x, −y). The class then repeats the process across the y-axis to find (x, y) → (−x, y). A short ARES video on reflecting points consolidates the discovery. Students close by measuring perpendicular distances from each vertex to each axis, confirming that image points are always equidistant from the line of reflection, and update the Driving Question Board with a new, more powerful question about non-axis mirror lines.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Intro to the coordinate plane]	Students are shown the Lake Victoria dawn photograph once more (projected on screen). The teacher places a transparent grid	Display the Lake Victoria photograph with the grid overlay. Say: 'Last lesson we found lines of symmetry by folding and by eye.	Predict–then–Test (Anticipation Guide variant): By committing predictions to paper before instruction, students activate prior	Collect or photograph predict sheets. Look for: (✓) students who correctly hold x constant and negate y for the x-axis reflection —

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	overlay on the image so that the waterline aligns exactly with the x-axis. Students are handed a half-sheet showing a simple boat polygon with vertices labelled A(2,3), B(5,3), C(6,1), D(4,0), E(1,1) plotted above the x-axis. Working individually for 3 minutes, students write their predictions: (a) Where will each vertex land after the water reflects the boat? Record your predicted image coordinates in the table provided. (b) Will the reflected image be larger, smaller, or the same size as the boat? (c) Will any coordinate value stay the same, and if so, which one? Students commit their predictions to paper before any instruction is given.	Today we are going to be far more precise — we are going to use coordinates to predict exactly where every point of the reflection lands.' Distribute the predict half-sheet. Say: 'You have 3 minutes. Do not discuss with your neighbour yet — I want YOUR prediction first.' Start a visible 3-minute timer. Circulate silently; note 2–3 students whose predictions show interesting or common misconceptions (e.g. sign errors on both coordinates, or swapping x and y). After 3 minutes, cold-call exactly 3 students using name cards: 'Amina, where did you predict vertex A will land? …Brian, same question — do you agree or disagree with Amina? …Zawadi, what is your reasoning?' Allow 10 seconds of WAIT TIME after each question before accepting an answer. Do NOT confirm or deny predictions — say: 'Hold that thought. Let us gather evidence.'	knowledge about symmetry and distance, create cognitive dissonance when results differ from predictions, and have a concrete record to compare against observations. This strategy makes the moment of discovery — when the rule emerges from the table — feel meaningful rather than arbitrary.	they are ready for extension; (△) students who negate both coordinates — a common misconception to address during Explain; (✗) students who have no systematic pattern — pair these students with a stronger peer during the Observe phase. Make a mental note of the 3 cold-called responses to revisit during Explain.
Observe Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Video (10 min): Students watch the ARES video 'Reflecting Points on the Cartesian Plane' and the companion clip 'Reflecting Objects across an Axis.' While watching, each student completes a structured viewing guide: they must pause-and-record every time a new coordinate pair (original → image) appears on screen, filling in a running table of at least 6 pairs. Part B — Hands-on Grid Activity (12 min): Each pair receives a large printed Cartesian grid (A3 size, axes labelled −8 to 8 on both x and y). Using the boat vertices from the predict sheet, they: (1) Plot and label original vertices A–E in pencil. (2) Fold the paper along the x-axis and hold it up to the	Before playing the video, say: 'As you watch, your only job is to be a data collector. Every time you see a point reflected, record the before and after coordinates in your table. I will stop the video twice to give you time to write.' Pause the video at the 2-minute mark and the 4-minute mark; each time say: 'Pause. 15 seconds — finish writing your current row.' After the video, transition to Part B: 'Now you are going to do exactly what the video showed — but with our Lake Victoria boat. The waterline IS the x-axis. Work with your partner.' Circulate during the fold-and-mark activity. Prompt pairs who are stuck: 'What happens to the height above the waterline	Data Collection Table + Physical Paper Folding: The fold-and-mark technique gives students a kinaesthetic, concrete experience of reflection before they abstract it into coordinate notation. Recording distances in millimetres grounds the algebraic rule in a physical measurement, linking the coordinate transformation to the KICD content statement that 'each point is the same perpendicular distance from the line of reflection as its corresponding point in the image.'	Walk the room during Part B and scan completed tables. Green indicator: the table shows x unchanged and y negated for x-axis reflection, and y unchanged and x negated for y-axis reflection. Amber indicator: one column is correct but the other is confused — prompt with: 'Look at your x-column only. What changed? What stayed the same?' Red indicator: coordinates are transposed or random — seat these students at the front for a teacher-led re-demonstration with two volunteers using the document camera. During Part C, check that measured distances match |y| for x-axis reflections; any discrepancy of more than 2 mm indicates a

	window (or classroom light) to see where the dots show through — mark those positions as A′–E′. (3) Unfold and record A′–E′ coordinates in a two-column table (Original | Image). (4) Repeat: re-fold along the y-axis, mark B″–E″, unfold, and record. Part C — Perpendicular Distance Check (5 min): Using a ruler, students measure the perpendicular (vertical) distance from each original vertex to the x-axis, and then the distance from its image to the x-axis, recording both in millimetres. They repeat for the y-axis reflection using horizontal distances.	when the boat is reflected into the water? Does it grow, shrink, or stay the same?' (WAIT 10 seconds.) For pairs who finish early, pose the extension: 'What do you notice about a point that sits exactly ON the x-axis — what happens to it?' During the perpendicular distance check, ask targeted pairs: 'If the boat is 3 units above the waterline, how far below the waterline is its reflection? What does that tell you about the rule?' (WAIT 12 seconds before cold-calling.)		plotting error to correct before Explain.
Explain Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students individually examine their completed two-column tables and answer three written prompts on their worksheet: (1) 'Write a rule in words: when you reflect a point across the x-axis, what happens to the x-coordinate? What happens to the y-coordinate?' (2) 'Write the same rule using coordinate notation: (x, y) → (?, ?).' (3) 'Write the equivalent rule for reflection across the y-axis.' After 4 minutes of individual writing, pairs share and agree on a single rule statement. The teacher then leads a whole-class 'Rule Gallery': four student pairs read their rules aloud; the class votes (thumbs up / thumbs sideways / thumbs down) on whether each statement is complete and precise. The teacher formalises the rules on the board: x-axis: (x, y) → (x, −y); y-axis: (x, y) → (−x, y). Students annotate their worksheets and then apply the rules to two new problems: (a) Reflect the maize-store polygon M(−3,4), N(0,4), P(0,1), Q(−3,1) across the y-axis. (b) A	After pairs have agreed on rule statements, choose 4 pairs strategically: one pair with the correct x-axis rule, one with the correct y-axis rule, one with a partial answer, and one with the 'negate both' misconception. Say to the class: 'I am going to ask four pairs to share. Your job is not to be polite — if you disagree, show me a thumbs sideways. If you think the rule is incomplete, thumbs down.' After each share, ask the class: 'Who can add to or correct this statement?' Use WAIT TIME of 10 seconds before accepting volunteers. When formalising: write the rules in colour on the board and say: 'Copy these exactly. These are not just tricks — they are describing what the water surface at Lake Victoria does to every single point of the boat, every single time.' For the application problems, cold-call 2 students to narrate their working aloud while plotting on the board grid. After the congruence verification, ask the whole class: 'Is	Rule Extraction from Data Table (Inductive Reasoning): Students move from specific coordinate pairs (data) to a general algebraic rule (abstraction). This mirrors the scientific practice of identifying patterns in evidence. The 'Rule Gallery' vote introduces peer accountability and requires students to evaluate mathematical language critically, deepening precision. Connecting the rule back to the phenomenon ('this is what the water does') anchors abstract algebra in the anchoring context.	Collect the written rule prompts (items 1–3 above). Specific observable indicators of mastery: student writes '(x, y) → (x, −y)' correctly with correct bracket notation; student can articulate WHY x stays the same ('the horizontal position does not change when you flip vertically'). Common error to flag: student writes '(x, y) → (−x, −y)' — this is a rotation, not a reflection; use a targeted follow-up question: 'If the boat is 2 metres to the right of the lighthouse on the shore, is its reflection still 2 metres to the right? Or does it move sideways into the water?' For the application problems, check that side lengths of the image polygon match originals to confirm congruence understanding.

	fisherman's net is modelled by points F(1,−2), G(4,−2), H(4,−5). Reflect it across the x-axis. Students verify that the image is congruent by measuring side lengths and confirming they match the original.	the image the same size as the original? How do you know — not by eye, but mathematically?' (WAIT 12 seconds.) Target answer: corresponding side lengths are equal, therefore the shapes are congruent.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students revisit the Driving Question Board displayed at the front of the room. The teacher reads aloud the main driving question: 'How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy — and how can we use that same idea to prove that two shapes are truly identical?' Each student receives two sticky notes. On the first (green): they write one thing today's lesson has helped them answer about the driving question. On the second (orange): they write one NEW question today raised that is not yet answered. Students place their notes on the DQB. The teacher reads 4–5 orange notes aloud and clusters them. The class discusses which cluster best captures today's boundary question. The teacher writes the agreed new question on the DQB in a dedicated 'Today's New Question' box: 'What if the mirror line is NOT the x-axis or the y-axis — how do we find where the reflection lands then?'	Point to the DQB and say: 'At the start of this unit we asked what rule the water is following. Let's see what we can now answer.' Read the driving question aloud slowly. Say: 'On your green note, write something you can now answer that you could not answer at the start of Lesson 1. On your orange note, write what you still cannot explain.' Give 2 minutes of silent writing. As students post notes, read 4–5 orange notes aloud without attribution: 'Someone has asked — what if the mirror line goes diagonally? Another person asks — does this work for curved shapes? Let's vote — which question should drive our next lesson?' After a show of hands, write the winning question on the board and say: 'This is exactly the question we will attack in Lesson 3. The answer requires a new tool. Keep that question in your mind tonight.'	Driving Question Board (DQB) Protocol: The DQB makes the progression of scientific and mathematical thinking visible over time. Green notes surface evidence of growing understanding; orange notes model the disposition of a mathematician — that answering one question always generates new, deeper questions. Clustering orange notes shows students that their individual confusions are shared, normalising productive struggle.	Read all green notes before Lesson 3. They reveal whether students can connect today's coordinate rules to the phenomenon (strong response: 'The x-axis IS the waterline, and the rule (x, y) → (x, −y) means the reflection sinks as far below as the boat rises above') or remain at a surface level ('I learned how to flip a shape'). Orange notes surface the most urgent misconceptions and curiosities to address in Lesson 3. If more than 40% of orange notes mention 'diagonal mirror lines,' prioritise that entry point for Lesson 3's predict phase.
Model Building Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English	Students open their cumulative 'Reflection Model' booklet (a folded A3 sheet that will grow across all 8 lessons). In Lesson 1 they drew a sketch of the Lake Victoria boat and marked lines of symmetry. Today they add a new layer: (1) They redraw the boat polygon on a	Distribute the cumulative model booklets. Say: 'Your model is your mathematical story of this phenomenon. Every lesson we add a new chapter. Today's chapter is about coordinates and rules.' Project a completed exemplar model page on the board	Cumulative Model Revision (Progressive Diagramming): By returning to the same physical booklet each lesson and literally adding to a growing diagram, students build a visual and written record of how their understanding deepens. The empty 'Lesson 3	Review collected booklets for: (✓) rule arrows include correct notation (x, y) → (x, −y) and (x, y) → (−x, y) alongside a verbal description; (✓) perpendicular distances are correctly labelled and match coordinate values (e.g., vertex at y = 3 is labelled 3 units

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Cartesian grid in their booklet, labelling all five original vertices and their x-axis images with coordinate pairs. (2) They annotate the diagram with two 'rule arrows': one showing (x, y) → (x, −y) for the x-axis, and one showing (x, y) → (−x, y) for the y-axis. (3) They draw a double-headed arrow between each original vertex and its image and label it with the measured perpendicular distance, writing the generalisation: 'distance from point to line of reflection = distance from image to line of reflection.' (4) In a 'Congruence Box' at the bottom of the page, they write: 'The image is congruent to the original because reflection preserves shape and size.' (5) They leave a clearly labelled empty space titled 'Lesson 3 addition: What happens when the mirror line is y = mx + c?'	(teacher-prepared). Say: 'I expect to see: the grid with labelled vertices, both rule arrows written in your own words, the perpendicular distances labelled, and the congruence statement. You have 6 minutes.' Circulate and check that students are annotating — not just drawing. Prompt: 'Your rule arrow is there, but what does it SAY? Write the mapping notation next to it.' At the 5-minute mark, say: 'You should be writing your congruence statement now.' Collect booklets (or photograph the page) to review before Lesson 3. Before closing, point to the empty 'Lesson 3 addition' box and say: 'That empty box is your homework in your mind tonight — think about what the rule might be when the mirror line is not an axis. Come ready to predict in Lesson 3.'	addition' box creates forward cognitive scaffolding — students know their model is incomplete by design, sustaining curiosity. Annotating perpendicular distances on the model cements the KICD content statement in a personally authored artefact.	from x-axis, image at y = −3 also labelled 3 units); (✓) congruence statement is written in own words and references 'same shape and size.' Flag for Lesson 3 small-group follow-up: any student whose model shows no annotation (diagram only) — they need a sentence-starter scaffold; any student who has labelled distances incorrectly — they need a targeted ruler-and-grid check at the start of Lesson 3.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before planning Lesson 3: (1) Did the paper-folding activity give students sufficient kinaesthetic grounding for the coordinate rules, or did some pairs struggle to align the fold precisely with the axis — would a different physical model (e.g., a mirror tile placed on the grid) work better? (2) When students voted on rule statements during the Rule Gallery, were dissenting thumbs (sideways/down) accompanied by mathematical reasoning, or was the vote social? If reasoning was thin, plan a 'convince me' protocol for Lesson 3. (3) Did the DQB orange notes cluster strongly around diagonal mirror lines, or did unexpected questions emerge (e.g., about curved reflections, or 3D)? Adjust Lesson 3's predict phase accordingly. (4) Were there students who finished the model-building phase early — did the extension question about points ON the axis generate productive discussion? (5) Most importantly: do students now articulate that the Lake Victoria reflection is not magic but mathematics — specifically, that the water surface is applying the rule (x, y) → (x, −y) to every point of the boat simultaneously? If not, open Lesson 3 with a direct re-connection to the phenomenon before moving to oblique mirror lines.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students watched the ARES videos on reflecting points and objects on the Cartesian plane, folded a printed grid along the x-axis and y-axis to physically locate reflection images of a boat polygon, and measured the perpendicular distances from each vertex to the line of reflection.
What did I learn?	Reflecting a point across the x-axis maps (x, y) → (x, −y); reflecting across the y-axis maps (x, y) → (−x, y). In both cases, each image point is exactly the same perpendicular distance from the line of reflection as the original point. The image is congruent to

	the original figure.
How does this explain the phenomenon?	The waterline in the Lake Victoria phenomenon acts as the x-axis: the water surface applies the rule $(x, y) \rightarrow (x, -y)$ to every point of the boat simultaneously, which is why the reflection is always the same size and always appears the same distance below the waterline as the boat sits above it.

LESSON 3 (40 minutes): Shifting the Mirror: Reflection Along Lines Parallel to the Axes	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students extend their understanding of reflection from the x- and y-axes to mirror lines of the form y = k and x = k, discovering how the same distance-preservation principle governs all reflections regardless of where the mirror line is placed.
Knowledge	Students will know that reflecting a point across the line y = k maps (x, y) to (x, 2k minus y), and reflecting across x = k maps (x, y) to (2k minus x, y); they will know that the perpendicular distance from a point to the mirror line equals the perpendicular distance from its image to the mirror line.
Skills	Students will be able to reflect points and polygons across lines of the form y = k and x = k on the Cartesian plane, state the coordinate transformation rule, plot the resulting image, and verify that image and object are equidistant from the mirror line.
Attitudes	Students will appreciate that a single underlying principle — equal perpendicular distance — unifies all reflections, developing confidence in generalising mathematical patterns from specific cases.
Key Inquiry Question	How do you apply reflection to real-life situations?
Purpose in Storyline	In Lesson 2 students discovered the axis-reflection rules and connected the x-axis to the waterline at Lake Victoria. A new question emerged: what if the mirror line is NOT the x- or y-axis? Lesson 3 answers that question by shifting the mirror line to any horizontal or vertical position, using matatu route landmarks as context. This extends the toolkit students need before tackling diagonal mirrors in Lesson 4 and eventually proving congruence in Lessons 6 and 7.
Safety Notes	No physical hazards. Ensure graph paper and rulers are shared equitably. If the classroom has a projector, reduce glare. Students with visual impairments should be seated near the front and given bold-printed grids.

B. LESSON OVERVIEW

The lesson opens by reconnecting to the Lake Victoria phenomenon: the teacher asks what would happen if the waterline were NOT at y = 0 but instead at y = 2 — would the reflection rule still hold? Students make written predictions. They then work through a structured activity in which they reflect named matatu-route landmarks (plotted as points and a simple polygon) across y = 2, y = minus4, x = 3, and x = minus1 on a Cartesian grid, recording image coordinates in a table and deriving the general rules. A sensemaking discussion surfaces the underlying distance principle and compares this lesson to the previous one. Students update their cumulative Cartesian models and add a new sticky note to the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan	Students are shown a projected Cartesian grid with a simple boat outline (from Lesson 2) plotted above the x-axis. The teacher then reveals a second grid on which a	Teacher displays the grid and says: In Lesson 2 we used the waterline — the x-axis — as our mirror. Today I am going to shift that mirror line upward to y = 2.	Predict-before-observe routine activates prior knowledge of the distance rule from Lesson 2. The shoulder-partner share surfaces misconceptions (most common:	Teacher circulates during the 90-second writing period and reads 4 to 5 student predictions over their shoulders, noting whether students are applying the axis rule

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	horizontal dashed line labelled y = 2 is drawn — and the boat outline has been moved so that some vertices sit above and some below that line. Students are given 90 seconds to write individual answers to two questions printed on their activity sheet: (1) If the mirror line is now y = 2 instead of y = 0, where do you predict the image of the point (1, 5) will land? (2) Will the image still be the same size as the object? Students then share predictions with a shoulder partner before a brief whole-class show-of-hands poll reveals the spread of predictions.	Imagine the lake surface has risen. Before we calculate anything, I want you to trust your instincts. Write your prediction for question 1 on your own — no talking yet. [WAIT TIME: 90 seconds of silent individual writing.] Now turn and share with your partner for 30 seconds. [WAIT TIME: 30 seconds.] Hands up — who predicts the image of (1, 5) lands at (1, minus1)? Who says (1, minus5)? Who says something else? Do NOT tell me why yet — we will test these shortly. The teacher records the three prediction categories on the board without evaluating them.	students incorrectly apply the axis rule and predict (1, minus5)) before the investigation begins, giving the teacher diagnostic information.	unchanged (common error), attempting a distance argument, or simply guessing. This informs which misconception to target during the Explain phase.
Observe Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive the Matatu Map Activity Sheet, which contains two pre-drawn Cartesian grids (axes from minus8 to 8 on both axes) and a supporting phenomenon context: a matatu company is designing a new route map for Kisumu and needs to create a mirror-image district layout for a planned expansion on the other side of a new road. Task 1 uses the line y = 2 as the road (mirror line). Four landmark points are given: the market at A(1, 5), the school at B(3, 4), the hospital at C(minus2, 6), and the bus terminus at D(4, 7). Students plot these, then physically count grid squares from each point perpendicularly to y = 2, continue the same number of squares on the other side, and plot the image points A-prime through D-prime. They record original and image coordinates in a table with columns: Point, Original coordinates, Distance above y=2, Image coordinates, Distance below y=2. Task 2 repeats the	Teacher says: Your matatu company needs to map out the mirror-image district. The road — your mirror line — is y = 2. Do NOT use any formula yet. Count the squares. For the market at (1, 5): how many squares above y = 2 is it? Count aloud with your partner. [WAIT TIME: 20 seconds.] Good — three squares above. So the image must be three squares below y = 2. What y-coordinate does that give? [WAIT TIME: 15 seconds.] Right — y = minus1. Fill in your table. Continue with B, C, D on your own. [WAIT TIME: 3 minutes for Tasks 1 and 2, circulating and prompting with: What is the perpendicular distance from this point to the mirror line? and Is your image on the correct side?] For Tasks 3 and 4 the mirror line is vertical — x = 3 or x = minus1. Pause before you start: will you count horizontally or vertically this time? Discuss with your partner. [WAIT TIME: 30 seconds.] Proceed — 2 minutes.	Concrete counting on the grid before any algebraic rule is introduced ensures students build the distance concept physically. The matatu-route context grounds the abstract plane in a familiar Kenyan transport scenario, making the activity meaningful and memorable.	Teacher checks tables during circulation. Key errors to flag (but not correct publicly yet): (a) students who subtract k from the y-coordinate instead of computing 2k minus y, (b) students who reflect vertically across a horizontal line correctly but then make the same error for a vertical mirror line. Teacher uses a clipboard checklist with student names to track who demonstrates correct distance counting versus who applies the axis rule mechanically.

	process for the polygon formed by A, B, C, D reflected across y = minus4. Task 3 reflects two points across x = 3. Task 4 reflects two points across x = minus1. Students join image points to form the image polygon after each task and check that corresponding sides look equal in length.			
Explain Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	After completing the activity sheet, students examine their completed coordinate tables and are guided to articulate the general rule. The teacher writes a partially completed table on the board for y = 2: original (1, 5) and image (1, minus1); original (3, 4) and image (3, 1); original (minus2, 6) and image (minus2, minus2); original (4, 7) and image (4, minus3). Students are asked to write in their notebooks: what operation was performed on the y-coordinate in each case? They share responses and the class works together to express the pattern as y-image = 2 times 2 minus y-original = 4 minus y-original, leading to the general rule for y = k: image is (x, 2k minus y). The teacher then repeats the algebraic generalisation for x = k, giving image (2k minus x, y). A brief class discussion connects this to the previous lesson: when k = 0 the rule gives (x, minus y) for y-axis reflection and (minus x, y) for x-axis reflection — exactly what students found in Lesson 2. Students record both rules in their notes with a worked example each.	Teacher points to the table on the board and says: Look at the y-coordinates only. Original is 5, image is minus1. Original is 4, image is 1. Original is 6, image is minus2. What is happening to each y-value? Write your answer as a calculation, not a word. [WAIT TIME: 60 seconds of individual writing.] Now share with the person behind you — 30 seconds. [WAIT TIME: 30 seconds.] Teacher cold-calls two students with contrasting answers. If a student says minus y plus 4, teacher says: Interesting — can you show me that works for all four rows? [WAIT TIME: 30 seconds.] If a student says 2 times 2 minus y, teacher says: Can you explain where the 2 times 2 comes from? [WAIT TIME: 30 seconds.] Teacher then formalises: The mirror line is y = k. Every point is at some distance d above it. Its image is at the same distance d below it. So image y = k minus d = k minus (y minus k) = 2k minus y. Write this in your notes. Now tell me: if k = 0, what does 2k minus y become? [WAIT TIME: 20 seconds.] Correct — minus y. So Lesson 2 was just a special case of today. Connection to phenomenon: If the waterline of Lake Victoria rose to y = 2 after a flood, every part of the boat would still reflect to the same	Algebraic generalisation is built from students own numerical data, not imposed top-down. The connection to k = 0 consolidates prior learning and reveals the elegance of a unified rule. The phenomenon callback maintains narrative coherence.	Teacher poses an exit check on the board: What is the image of (minus3, 1) under reflection in y = minus4? Students write the answer on a sticky note and hold it up. Teacher scans for (minus3, minus9). Common error: (minus3, minus5) indicates the student subtracted 4 instead of applying 2k minus y with k = minus4. Teacher notes names for follow-up at the start of Lesson 4.

		perpendicular distance below the new waterline. The reflection rule shifts, but the principle — equal distance — never changes.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board. The teacher reads aloud the sticky note added at the end of Lesson 2: What if the mirror line is NOT the x- or y-axis? Students are given one green sticky note each. On it they write one sentence explaining how today answered that Lesson 2 question, and one new question that today raises for them. Students post their green notes on the DQB. The teacher reads four or five aloud and clusters them. Likely emerging new questions include: What if the mirror line is diagonal — like y = x? Does the rule change completely? Can two different mirror lines produce the same image? How do we know the image is always exactly the same size as the object no matter where the mirror line is?	Teacher says: At the end of Lesson 2, someone asked — and I am reading their own words — What if the mirror line is not the x- or y-axis? Look at your green sticky note. In one sentence, has today answered that question? Write yes and explain, or write not fully and explain what is still missing. Then write the new question today has opened for you. [WAIT TIME: 90 seconds of silent writing.] Post your note. Teacher reads selected notes aloud and says: I notice several of you are asking about diagonal mirror lines. That is exactly where we are heading in Lesson 4. I also notice some of you are asking how we are sure the image is always the same size. We will give that question a very precise mathematical answer in Lesson 6. Keep it on the board — it is a powerful question.	The DQB update creates explicit continuity between lessons and models scientific practice of revising questions in light of new evidence. Clustering emerging questions previews the rest of the storyline and gives students a sense of agency in the inquiry.	Teacher photographs the updated DQB after class. Quality of student-written sentences on green notes reveals whether students can articulate the distance principle in their own words, providing a formative writing sample for the teacher to review before Lesson 4.
Model Building Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12	Students open their cumulative Cartesian model — a large grid begun in Lesson 2 on which they recorded the x-axis and y-axis reflections of a boat outline. Today they add two new elements to the same model: (1) they draw a dashed horizontal line at y = 2 and plot the boat outline reflected across it, labelling the image in a different colour and annotating the equal perpendicular distances with small arrows; (2) they write the two new rules in a legend box at the bottom of the model: Reflection across y = k gives image (x, 2k minus y); Reflection across x = k	Teacher says: Your cumulative model is your scientific record of everything we have discovered so far about reflection. Add today's two new mirror lines and rules. Use a different colour for each mirror line so you can distinguish them at a glance. When you write the annotation for equal perpendicular distances, use the word equal and a measurement — do not just draw an arrow without a number. [WAIT TIME: 4 minutes for model updating, teacher circulates.] Teacher prompts early finishers: Can you prove algebraically, using the rule, that	The cumulative model functions as a visible thinking record that grows across all 8 lessons, making the progression of ideas concrete and reviewable. Adding each new case to the same grid allows students to visually compare all reflection types and builds towards the full conceptual model needed for congruence.	Teacher collects or photographs 6 models (two from each ability group) to check whether the annotations demonstrate conceptual understanding of equal perpendicular distances rather than mere mechanical plotting. Feedback is given as written comments at the start of Lesson 4.

🔍 Search ARES for similar readings	gives image (2k minus x, y). Students also add a brief note connecting to the phenomenon: If the Lake Victoria waterline is at height k, the reflection rule becomes y-image = 2k minus y-object. The model now shows three mirror lines: y = 0, y = 2, and optionally x = 3, illustrating how the same boat looks different depending on where the mirror line is placed, yet the image always remains the same size.	the distance from (1, 5) to y = 2 equals the distance from (1, minus1) to y = 2? Write it as a subtraction. [WAIT TIME: 60 seconds.] After 4 minutes teacher asks one student to hold up their model and narrates: Notice how the boat shrinks and grows in your drawings — wait, does it? [Pause.] No, it does not. Every image is exactly the same size. That is our big idea in Lesson 6 — and it has a name: congruence.		

D. TEACHER REFLECTION

After the lesson, consider: Did students readily transfer the distance-counting strategy from y = 0 to y = 2, or did a significant number revert to the axis rule? Were students able to articulate the 2k minus y formula in their own words, or did they memorise it without understanding? Did the matatu context feel authentic and engaging, or did students treat it as a superficial wrapper? Which new DQB questions suggest genuine conceptual progress versus surface-level curiosity? What does the sticky-note exit check reveal about readiness for diagonal reflections in Lesson 4? Were students with lower prior attainment able to succeed on the counting tasks even if the algebraic rule remained unclear — and how will you bridge that gap?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Using the matatu district map on the Cartesian plane, students plotted landmark points and counted perpendicular grid squares to find their images across y = 2, y = minus4, x = 3, and x = minus1, recording original and image coordinates in a table.
What did I learn?	Reflecting a point across y = k maps (x, y) to (x, 2k minus y); reflecting across x = k maps (x, y) to (2k minus x, y). In both cases the image is the same perpendicular distance from the mirror line as the original, and the reflection for k = 0 is the special case discovered in Lesson 2.
How does this explain the phenomenon?	If the waterline of Lake Victoria were at any height k above the standard axis — for example after seasonal flooding — the boat reflection would still land at an equal perpendicular distance below that new waterline, because the equal-distance rule works for any horizontal mirror line, not just y = 0.

LESSON 4 (80 minutes): Diagonal Mirrors: Reflection Along y = x and y = −x

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students extend their understanding of reflection to diagonal mirror lines — specifically y = x and y = −x — discovering the coordinate rules (x, y) → (y, x) and (x, y) → (−y, −x), and confirming that the resulting image is always congruent to the original figure.
Knowledge	Students will know that: (1) reflecting across y = x swaps the x- and y-coordinates of every point; (2) reflecting across y = −x negates and swaps the coordinates; (3) the line of reflection is always given in the form y = mx + c; (4) each point in the starting figure is the same perpendicular distance from the line of reflection as its corresponding point in the image; (5) the image produced by any reflection is congruent to the original figure.
Skills	Students will be able to: (1) reflect a point and a polygon across y = x and y = −x on a Cartesian plane; (2) apply and verify the coordinate rules (x, y) → (y, x) and (x, y) → (−y, −x); (3) measure perpendicular distances from the line of reflection to confirm equidistance; (4) identify the line of reflection when given only an object and its image; (5) use correct geometric notation for image points (prime notation: A′, B′, C′).
Attitudes	Students will appreciate the elegance of mathematical rules that describe physical phenomena (such as tile patterns and water reflections), develop persistence when grappling with productive misconceptions, and collaborate respectfully during paired and group investigations.
Key Inquiry Question	How do you apply reflection to real-life situations?
Purpose in Storyline	In Lessons 1–3 students established the reflection rules for horizontal (y = c) and vertical (x = c) mirror lines and connected them to the Lake Victoria phenomenon. Lesson 4 advances the inquiry by introducing diagonal mirror lines, inspired by the diagonal tile-symmetry patterns observed in Mombasa's old-town architecture. Students discover that the water-surface 'rule' — equal perpendicular distance on both sides — still holds, but now produces a coordinate-swap rather than a sign-change, deepening their model of what reflection really does and pushing the class closer to a formal definition of congruence.
Safety Notes	No significant safety concerns. Ensure rulers and compasses are handled carefully. If printed grid sheets are used, ensure students do not obstruct each other's workspace during the paired activity.

B. LESSON OVERVIEW

Lesson 4 of 8 in the evidence-gathering sequence. Students begin by revisiting the Driving Question Board and briefly recapping the reflection rules discovered in Lessons 1–3 (reflections across horizontal and vertical lines). The lesson then pivots to a supporting phenomenon: the geometric tile patterns on the walls of old buildings in Mombasa's Old Town, where diagonal lines of symmetry are visible. Through a full Predict–Observe–Explain cycle, students investigate reflections across y = x and y = −x, discover the coordinate transformation rules, address the productive misconception that 'diagonal reflections look like rotations', and update the DQB with new evidence. The lesson closes with a cumulative model revision in which students annotate their running 'Lake Victoria reflection model' to include diagonal mirror lines. All activities reinforce the core insight: regardless of the orientation of the mirror line, every reflected image is congruent to its original — same shape, same size, equidistant from the line of reflection.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cofunction Identities and Reflection (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students are shown a projected image of a decorative tile panel from Mombasa's Old Town fort walls. The tile panel has a clear diagonal line of symmetry running from bottom-left to top-right. A second slide shows a simple triangle plotted on a Cartesian grid with the line y = x drawn in a dashed red colour — but the image (reflection) is hidden. Working individually in their exercise books, students write a prediction: 'Where will each vertex of the triangle land after it is reflected across y = x? Write the predicted coordinates.' After 3 minutes of silent individual thinking, students share predictions with a shoulder partner for 2 minutes. The teacher cold-calls 4 pairs (rotating around the room) to share their predicted coordinates aloud. Predictions are recorded on a shared class 'Prediction Anchor Chart' on the board — including wrong predictions — so they can be revisited in the Explain phase.	1. Display the Mombasa tile image and ask: 'Look at this tile panel from the walls of Fort Jesus in Mombasa. Where do you see a line of symmetry? Point to it.' Allow 10 seconds WAIT TIME before cold-calling 3 students by name. 2. Transition to the grid slide: 'We have reflected across horizontal and vertical lines. Today our mirror line is diagonal — it is the line y = x. Without calculating anything yet, predict: if I reflect triangle ABC across this diagonal line, where will the vertices land?' 3. Circulate silently for 3 minutes while students write individual predictions. Do NOT confirm or deny any prediction at this stage. 4. After pair-share, cold-call exactly 4 pairs; for each, ask: 'What reasoning did you use to make that prediction?' Record all predicted coordinate sets on the board in a two-column table labelled 'Original | Predicted Image'. 5. Say: 'Hold on to your predictions — we are going to test them right now with evidence.'	Prediction Anchor Chart — by making predictions public and recorded before observation, students create cognitive dissonance that they are motivated to resolve. This strategy surfaces prior knowledge (and misconceptions) and gives students a personal stake in the evidence-gathering that follows.	Observable indicator: Are students attempting to apply a coordinate rule, or are they drawing visually/estimating? Listen for students who say 'I just flipped it over the line' vs. those who attempt a coordinate approach. Note names of students who predict (y, x) correctly — they may be ready for extension. Flag students who predict no change or a rotation — they need targeted support in Phase 3.
Observe Phase 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cofunction Identities and Reflection]	Students carry out a two-part structured investigation using printed A3 grid sheets (or exercise-book grids if A3 is unavailable). PART A — Reflecting across y = x: Each student plots triangle ABC with A(1, 3), B(4, 2), C(3, 5) and the line y = x on their grid. They then use a ruler to draw the perpendicular from each vertex to the line y = x, measure the distance from vertex to line, and mark the image point at the same	1. Distribute grid sheets and say: 'This is your evidence-collection phase. Your job is to be precise — use a sharp pencil, a ruler, and measure carefully. The mirror line y = x is the line where x and y are equal — plot a few points to draw it first: (0,0), (1,1), (2,2), (3,3).' 2. After students draw the line, ask: 'What angle does y = x make with the x-axis?' WAIT 10 seconds, cold-call 2 students. Confirm: 45°. 3. Circulate continuously. When you spot a student drawing a	Structured Observation Table with 'Pattern I Notice' column — this is a data-recording strategy that deliberately separates raw data (coordinates) from pattern recognition, scaffolding students toward generalisation without telling them the rule. The gallery walk adds a peer-verification layer and surfaces discrepancies for class discussion.	Observable indicator: Check that perpendicular distances from object to line equal perpendicular distances from image to line — this is the core geometric property. Collect 4–5 exercise books during the gallery walk and quickly scan the results tables. Look for: (a) correct perpendicular construction technique, (b) correct image coordinates in the results table, (c) any student who has already written the swap-rule in the 'Pattern I Notice' column (high

[(Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	perpendicular distance on the other side. They connect the image points to form triangle A′B′C′ and record the coordinates of A′, B′, C′ in a results table. PART B — Reflecting across y = −x: Students repeat the process for triangle PQR with P(2, −1), Q(5, −3), R(3, −5) reflected across y = −x. They again measure perpendicular distances and record image coordinates P′, Q′, R′. Students then complete a structured observation table: Original Coordinates | Image Coordinates | Pattern I Notice. The class pauses for a 5-minute 'gallery walk' around 4 posted large-grid examples (prepared by the teacher on manila paper) showing the same constructions at A1 size, allowing students to self-check their work before moving to explanation.	perpendicular incorrectly (i.e., not at 90° to the line), kneel to their level and ask: 'Show me how you checked that this line meets y = x at a right angle.' Do not correct — prompt. 4. After Part A, pause the class: 'Pause pencils. Look at your results table. What do you notice about the relationship between the original coordinates and the image coordinates? Write one sentence.' WAIT 15 seconds. 5. Cold-call 3 students. Record their 'noticings' verbatim on the board — do not evaluate yet. 6. Direct students to Part B: 'Now try y = −x. This line passes through (0,0), (1,−1), (2,−2). Plot it as a dashed blue line.' 7. During gallery walk, prompt students: 'Find one poster where the image looks like what you drew and one that surprises you. Write a question on a sticky note and attach it to the poster.'		readiness). Flag any student whose image triangle appears rotated or scaled — address in Phase 3.
Explain Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students transition from data to generalisation. First, they complete a 'Coordinate Rule Discovery' structured worksheet individually (5 minutes): they fill in the blanks — 'When I reflect (x, y) across y = x, the image is (__, __). When I reflect (x, y) across y = −x, the image is (__, __).' — using their results table as evidence. After individual work, students join groups of 3 to compare rules and reach consensus. Each group writes their agreed rules on a mini-whiteboard (or a half-sheet of paper) and holds it up. The teacher addresses the lesson's key productive misconception: several students will likely claim the diagonal reflection 'looks like a	1. Begin whole-class discussion: 'Let's look at the patterns you found. If the original point is (1, 3), what did you get as the image across y = x?' Cold-call 3 students. Write on board: (1, 3) → (3, 1). Repeat with (4, 2) → (2, 4) and (3, 5) → (5, 3). Ask: 'What is the rule? Describe it in one sentence.' WAIT 15 seconds. 2. After students articulate the swap: 'Excellent. Mathematically we write: reflection across y = x maps (x, y) → (y, x). Write this in a box in your notes.' 3. Repeat for y = −x: guide students to (2, −1) → (1, −2), (5, −3) → (3, −5), etc., and elicit (x, y) → (−y, −x).	Claim–Evidence–Reasoning (CER) Framework — students make a claim (the coordinate rule), cite evidence (their results table), and provide reasoning (the perpendicular-distance property and the coordinate pattern). Revisiting the Prediction Anchor Chart makes the learning arc explicit: students literally see in colour how their thinking changed from prediction to evidence-supported explanation.	Observable indicator: Can students correctly state both rules in symbolic form without looking at their notes? During mini-whiteboard display, check for: (a) correct notation (y, x) and (−y, −x), (b) use of arrow notation (x, y) → (y, x), (c) ability to apply the rule to a new point not used in the investigation. Students who conflate reflection with rotation should be asked: 'Is the orientation of the shape the same after a reflection? After a rotation?' — a correct answer distinguishes the two transformations.

	rotation.' The teacher uses a direct comparison activity — overlaying a rotation of 90° on the same triangle and showing it produces a different image — to make the distinction vivid. Students then apply both rules to two new polygons: a quadrilateral DEFG and a real-world 'boat outline' polygon (5 vertices) inspired by the Lake Victoria dhow, connecting back to the anchoring phenomenon. Finally, students revisit the Prediction Anchor Chart from Phase 1 and publicly identify which predictions were correct, which were partially correct, and which were misconceptions, annotating the chart in a different colour.	4. Address the misconception directly: 'I heard some of you say this looks like a rotation. Let's test that.' Draw triangle ABC on the board. Show a 90° rotation image alongside the y = x reflection image. Ask: 'Are these the same point? Point A rotated 90° lands at ___, but A reflected across y = x lands at ___. Are they the same?' Cold-call 2 students. Conclude: 'Reflection and rotation are different transformations — even when they look similar at first glance.' 5. For the dhow polygon activity, say: 'Here is a simplified outline of a Lake Victoria fishing dhow plotted on a grid. Reflect it across y = x. This is the boat; y = x is the waterline at a diagonal. What do you notice about the image?' 6. Return to Prediction Anchor Chart: 'Using a green pen, put a tick next to every prediction that matched our rules. Circle the ones that were misconceptions. What does this tell us about how our thinking has grown since the start of the lesson?' 7. Pose the congruence link: 'Measure the side lengths of your original triangle and your image triangle. What do you find?' WAIT 10 seconds. Elicit: 'They are equal.' Conclude: 'The image is congruent to the original — same shape, same size. This is true for any reflection, across any mirror line.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Properties of]()	Students return to the class Driving Question Board displayed at the front of the room. The DQB has been running since Lesson 1 and contains sticky-note questions, evidence cards, and	1. Direct students to the DQB: 'Take 3 minutes with your partner — write your green sticky note and your yellow sticky note. Be specific; use coordinates or the words of the rule.' Circulate and	Driving Question Board as Living Document — the DQB functions as a cumulative public record of the class's growing understanding. By physically connecting today's evidence to prior questions with	Observable indicator: Are students' 'We now know…' statements specific and evidence-based (include the symbolic rule and/or the congruence conclusion) or vague ('we learned about

[Equality and Congruence Principles - Basic](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cofunction Identities and Reflection (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	'We now know…' statements from previous lessons. Working in pairs, students: (1) write one new 'We now know…' statement on a green sticky note based on today's evidence (e.g., 'We now know that reflecting across y = x swaps the coordinates, and the image is always congruent to the original'); (2) write one new question that today's lesson raised for them on a yellow sticky note (e.g., 'How would we find the mirror line if we were only given the object and its image?'); (3) identify one sticky note already on the DQB from a previous lesson that today's evidence helps to answer, and draw an arrow connecting them with a red marker. The teacher facilitates a 5-minute whole-class DQB share: 3 pairs read their 'We now know…' statements aloud; 2 pairs share their new questions. The teacher explicitly connects today's lesson to the overarching Driving Question: 'We are getting closer to answering how reflection proves two shapes are truly identical — what new evidence are we adding today?'	prompt vague statements: 'Can you add the actual rule in symbols?' 2. After 3 minutes, cold-call 3 pairs to read their green notes aloud. Post them on the DQB under the heading 'Lesson 4 Evidence'. 3. Cold-call 2 pairs to share their yellow (question) notes. Ask the class: 'Has anyone started thinking about how to answer this question?' WAIT 10 seconds before taking responses. Post yellow notes under 'New Questions'. 4. Point to the Driving Question on the board and read it aloud: 'How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy — and how can we use that same idea to prove that two shapes are truly identical?' 5. Ask: 'Which part of this question did today's lesson help us move closer to answering?' Cold-call 2 students. Listen for: 'We can now say the image is congruent to the original for diagonal reflections too — it is always the same size.' 6. Preview Lesson 5: 'Next lesson we will look at the reverse problem — you will be given only the object and the image and you will have to find the mirror line. Your yellow sticky-note questions are exactly what we will investigate.'	arrows, students experience the iterative, evidence-building nature of scientific and mathematical inquiry. The act of writing 'We now know…' statements consolidates declarative knowledge and makes learning progress visible.	reflections')? The quality of the green sticky notes is a reliable proxy for conceptual understanding. Collect 4–5 yellow notes to inform Lesson 5 planning — high-quality questions (e.g., 'What if the mirror line does not pass through the origin?') indicate readiness for extension.
Model Building Phase 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic](#)	Students retrieve their 'Lake Victoria Reflection Model' — a cumulative annotated diagram they have been building since Lesson 1. The model currently shows a dhow above a horizontal waterline and its reflection, annotated with the Lesson 1–3	1. Say: 'Take out your Lake Victoria Reflection Model. Look at what you have built over the last three lessons. Today you are adding the diagonal mirror lines to your model.' Allow 1 minute for students to review their existing models.	Cumulative Model Revision — by returning to and adding to the same running model every lesson, students experience knowledge as building and interconnected rather than isolated. The Model Gallery Pause uses peer models as additional instructional resources,	EXIT TICKET (3 minutes, individual, written): 'Point A(−3, 5) is reflected across y = −x. (a) Write the coordinates of A′. (b) Explain in one sentence how you know the image is congruent to the original. (c) On the small grid below, sketch the line y = −x, plot A, and plot A′.'

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	rules for horizontal and vertical mirror lines. In Lesson 4, students add a new panel to their model: they sketch a coordinate grid, draw the line y = x, and show a simplified boat outline being reflected across it. They annotate: (1) the coordinate rule (x, y) → (y, x); (2) an arrow from one vertex showing the perpendicular distance to the line and the equal distance on the other side; (3) the statement 'Image is congruent to original — same shape, same size'; (4) a note connecting to the Mombasa tile phenomenon — 'This is how diagonal tile symmetry works too.' Students also add a second small panel for y = −x with the rule (x, y) → (−y, −x). The lesson closes with a 'Model Gallery Pause' — students hold up their models and the teacher selects 2–3 to display under the document camera, narrating what strong annotations look like. Students have 2 minutes to improve their own model based on what they see.	2. Prompt: 'In your new panel, I need to see four things — the coordinate rule in symbols, the perpendicular distance annotation, the congruence statement, and your Mombasa tile connection. You have 8 minutes.' 3. Circulate and narrate what you see: 'I can see Achieng has drawn a clear perpendicular from vertex B to the line y = x and measured equal distances on both sides — well done.' Use student names to make praise specific and public. 4. After 8 minutes, conduct the Model Gallery Pause under the document camera. For each displayed model, ask the class: 'What is strong about this annotation? What could be added?' Cold-call 3 students. 5. Give 2 minutes for model improvement. 6. Close with exit ticket (see Formative Assessment below): pose the question verbally and give students 3 minutes to respond in writing before the bell. 7. Final connection to phenomenon: 'When a fisherman at Lake Victoria sees his dhow reflected at dawn, he sees a congruent image. We now have the mathematical rule for why that is true — whether the mirror line is horizontal, vertical, or diagonal. What is the one rule that never changes, no matter which mirror line we use?' WAIT 15 seconds. Cold-call 2 students. Listen for: 'The perpendicular distance from object to mirror line equals the perpendicular distance from image to mirror line.'	leveraging social learning and giving students concrete exemplars of quality mathematical communication.	Observable indicators: (a) correct application of rule → A′(−5, 3); (b) student references equal perpendicular distances or equal side lengths; (c) correct positioning of both points relative to y = −x. Students who answer all three correctly are ready for Lesson 5 extension (finding the mirror line from object and image). Students who apply the wrong rule (e.g., y = x rule) need a brief 1-on-1 clarification at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students successfully distinguish between reflection and rotation during the Explain phase? If the misconception persisted, consider starting Lesson 5 with a side-by-side comparison task. (2) How precise were students' perpendicular-distance constructions in the Observe phase? If many students struggled with the 45° perpendicular, build a 5-minute ruler-and-set-square construction drill into the start of Lesson 5. (3) Review the exit tickets before Lesson 5: tally how many students correctly applied $(x, y) \rightarrow (-y, -x)$ for $y = -x$ versus those who applied the $y = x$ rule. If more than 30% of students confused the two rules, open Lesson 5 with a brief paired correction activity. (4) Were the DQB yellow sticky-note questions rich enough to drive Lesson 5's inquiry into finding the mirror line from object and image? If questions were superficial, model a richer question format at the start of Lesson 5. (5) Did the Mombasa tile supporting phenomenon successfully motivate the diagonal mirror line, or did it feel disconnected from the Lake Victoria anchoring phenomenon? If disconnected, strengthen the bridge by explicitly asking students: 'If you tilt the Lake Victoria waterline to 45°, which reflection rule would apply?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that when triangle ABC was reflected across the line $y = x$ on a Cartesian grid, each vertex landed at a point where the x- and y-coordinates were swapped — for example, A(1, 3) mapped to A′(3, 1). They also observed that when reflecting across $y = -x$, both coordinates were swapped and negated — for example, P(2, −1) mapped to P′(1, −2). In both cases, students measured and confirmed that each vertex was the same perpendicular distance from the mirror line as its corresponding image vertex.
What did I learn?	Students learned that: (1) reflecting a point across $y = x$ follows the rule $(x, y) \rightarrow (y, x)$; (2) reflecting a point across $y = -x$ follows the rule $(x, y) \rightarrow (-y, -x)$; (3) the mirror line in any reflection is always equidistant from object and image (each point and its image are the same perpendicular distance from the line of reflection); (4) the image produced by reflection is always congruent to the original figure — same shape, same size; (5) diagonal reflections are not the same as rotations, even though they can look similar.
How does this explain the phenomenon?	Students explained their observations using the Claim–Evidence–Reasoning framework: they claimed the coordinate rules for $y = x$ and $y = -x$, cited their measured coordinate data as evidence, and reasoned that the rules work because the perpendicular-distance property of reflection forces an exact coordinate swap (for $y = x$) or a negate-and-swap (for $y = -x$). They connected this back to the Lake Victoria phenomenon by noting that no matter what orientation the mirror line takes — horizontal, vertical, or diagonal — the reflected boat image is always congruent to the original, because the fundamental property of reflection (equidistance from the mirror line) never changes.

LESSON 5 (80 minutes): Reverse Engineering the Mirror: Finding the Equation of the Mirror Line	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students determine the equation of a mirror (reflection) line by constructing perpendicular bisectors of object-image point pairs plotted on a coordinate grid, connecting this procedure to the anchoring phenomenon of the hidden waterline at Lake Victoria.
Knowledge	Each point in the original figure is the same perpendicular distance from the line of reflection as its corresponding point in the image. The line of reflection can always be expressed in the form y = mx + c.
Skills	Plotting object-image pairs on a coordinate grid; constructing perpendicular bisectors of line segments joining corresponding points; calculating midpoints and gradients; writing the equation of the mirror line in the form y = mx + c.
Attitudes	Curiosity and persistence in working backwards from evidence to find a hidden rule; appreciation that mathematics can reveal structure that is invisible to the eye.
Key Inquiry Question	How do you determine the line of symmetry of a figure? How do you apply reflection to real-life situations?
Purpose in Storyline	In earlier lessons students mapped reflections given a known mirror line. Today they reverse the process: given only an object and its image, they reconstruct the hidden waterline — just as a fisherman on Lake Victoria might ask, 'If I can see both the boat and its reflection but the shore is in fog, where exactly is the water surface?' This reversal deepens understanding of the perpendicular-distance property and introduces the equation of the mirror line as a formal mathematical object.
Safety Notes	Compasses have sharp points — remind students to place the compass tip carefully on the paper and keep fingers clear of the point when adjusting. Rulers and set squares should be handled with care. No other safety concerns for this lesson.

B. LESSON OVERVIEW
Lesson 5 of 8 in the evidence-gathering sequence on Congruence. Students open by revisiting the anchoring phenomenon with a new puzzle: the waterline of Lake Victoria is hidden by morning mist — can they locate it using only the boat and its mirror image? After a brief DQB review, students watch a short video on finding mirror lines, then undertake a hands-on grid activity in which they are given pre-plotted object-image pairs, construct perpendicular bisectors of corresponding point-pairs, identify the midpoints, calculate the gradient of the mirror line, and write its equation in the form y = mx + c. Whole-class sensemaking consolidates the perpendicular-distance property. The lesson closes with DQB updates and a model-revision task in which students annotate their cumulative reflection model with the newly discovered equations.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan	Students are shown a projected image: a simplified coordinate grid with a boat shape (polygon ABCD) and its mirror image (A'B'C'D') already plotted — but the mirror	Display the projected grid (boat + image, no mirror line). Read the phenomenon hook aloud with energy: 'The mist is on the water — you cannot see the shore. You	Think-Pair-Predict: Students first commit individually in writing (reducing anchoring bias), then articulate reasoning to a peer, then hear contrasting predictions	Teacher scans written predictions for two observable indicators: (1) Does the student reference the midpoint or equal distance property learned in Lesson 3? (2)

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	line is deliberately hidden (erased/not drawn). The phenomenon hook is re-stated aloud by the teacher: 'It is dawn on Lake Victoria. The mist is sitting right on the water surface. You can see the fishing boat above and its perfect reflection below — but you cannot see the actual waterline. Could you figure out exactly where the water surface is, using only the mathematics we have learned?' Students individually write a prediction in their exercise books: (a) Describe in words how you would try to locate the hidden mirror line. (b) Sketch on the mini-grid provided where you think the mirror line sits. (c) Write one question you still have about the process. Students then share predictions with a shoulder partner (Think-Pair) before selected pairs report to the class.	have only the boat and its shadow in the water. Where is the waterline?' Allow 10 seconds of silent viewing before issuing the written prediction task. Circulate for 4 minutes as students write; note 2–3 interesting or contrasting predictions to use in cold-call. After Think-Pair (2 minutes), cold-call exactly 3 pairs using a name-stick or seating chart: one pair likely to say 'find the middle between the points,' one pair likely to mention perpendicular lines, and one pair with an unexpected idea. After each response, ask: 'What evidence from our earlier Lake Victoria lessons makes you think that?' Do NOT confirm or correct at this stage — record predictions on the left-hand side of the DQB under 'What we think we know (Lesson 5).' Close predict phase by saying: 'Keep your prediction visible — we will return to it at the end of the lesson to see how close you were.'	publicly. This surfaces prior knowledge about midpoints and perpendicularity from Lessons 1–4 and creates a 'prediction gap' that motivates the observation phase.	Does the student use the word 'perpendicular' or draw a right-angle symbol? Students who do neither are flagged for targeted questioning during the activity phase. Teacher records 3–4 names for follow-up.
Observe Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Video (approx. 10 min): Students watch a teacher-curated video on 'Finding the Mirror Line Given Object and Image.' As they watch, they complete a structured viewing guide with three sentence starters: (1) 'The mirror line passes through the ______ of each segment joining a point to its image because…' (2) 'The mirror line is ______ to each of those segments because…' (3) 'To write the equation y = mx + c, I need to find…' Students pause at two teacher-signalled moments to discuss sentence starters with their partner. Part B — Hands-on Grid Activity (approx. 20 min): Each pair receives Activity Sheet 5 — a pre-printed coordinate grid	Before starting the video, distribute viewing guides and say: 'You are watching this video like a detective — you are looking for the mathematical rule the mirror line must obey. Pause your thinking at my signal.' Play the video. Pause at the midpoint-construction moment and give 15 seconds WAIT TIME before asking: 'Tell your partner: why does the perpendicular bisector always pass through the midpoint — what does that have to do with equal distances?' Cold-call 2 pairs. Resume video. After the video, say: 'Compare your sentence starters with your partner. Agree on one best answer for each.' Transition to Part B: hand out	Structured Viewing Guide with Strategic Pauses: Sentence starters require students to process the video actively rather than passively. Pausing at key moments and requiring partner discussion mirrors the 'scientific practice of obtaining, evaluating, and communicating information' (NGSS SEP 8). The hands-on construction activity embeds NGSS SEP 3 (Planning and Carrying Out Investigations) — students follow a structured procedure, collect data (midpoint coordinates, gradients), and use it to produce a mathematical claim (the equation).	Circulating teacher checks two specific indicators on activity sheets: (1) Are perpendicular bisector arcs drawn correctly (arcs crossing on both sides of the segment)? — if not, pause the student and demonstrate with the board compass. (2) Has the student correctly substituted the midpoint coordinates into y = mx + c to solve for c? — check arithmetic. Teacher also asks any flagged student from Phase 1 to verbally explain Step 2 before moving on.

	(−10 to 10 on both axes) showing three separate object-image pairs: (i) Triangle KLM and its image K'L'M' reflected in an oblique mirror line; (ii) Quadrilateral PQRS and its image P'Q'R'S' reflected in a horizontal mirror line; (iii) Line segment EF and its image E'F' reflected in a vertical mirror line. Students use ruler and compass to: Step 1 — Draw the segment joining each object point to its image point (e.g., KK', LL', MM'). Step 2 — Construct the perpendicular bisector of each segment (compass arcs, ruler). Step 3 — Identify the midpoint of each segment by reading its coordinates from the grid. Step 4 — Use two midpoints to calculate the gradient of the mirror line. Step 5 — Substitute one midpoint into y = mx + c to find c and write the full equation. Students record all working on the activity sheet. Real-world link printed at the top of the activity sheet: 'You are a hydrographer for the Kenya Meteorological Department, using aerial photographs of Lake Victoria to locate the exact shoreline — but the waterline is obscured by morning mist. You can see the boats and their reflections. Find the waterline.'	Activity Sheet 5 and read the hydrographer scenario aloud. Circulate during the activity, spending approximately 3 minutes per cluster of 4 desks. Use targeted questions for flagged students: 'Show me the segment you drew from K to K'. Now — what is its midpoint?' For students who finish early, pose the extension: 'Could you find the mirror line using only ONE object-image pair? Why or why not?' At the 15-minute mark of Part B, issue a 5-minute warning and ask all pairs to ensure they have written at least one full equation.		
Explain Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and	Whole-class sensemaking debrief (approx. 15 min). The teacher projects a clean version of Activity Sheet 5, Item (i) — Triangle KLM — on the board. Three volunteer pairs (one per item) come to the front in turn and: (a) point to the perpendicular bisectors they drew; (b) state the midpoint coordinates they read; (c) write the gradient calculation on the board; (d) write and read aloud the equation of the	Select the three presenting pairs deliberately — choose one strong pair, one middle pair, and one pair that made a correctable error, to model how errors advance understanding. Before each presentation, tell the class: 'Your job is to listen for the Evidence step — can they prove the line passes through all three midpoints?' After each presentation, give 10 seconds	Claim–Evidence–Reasoning (CER) Framework: CER is an explicit argumentation structure (NGSS SEP 7 — Engaging in Argument from Evidence) that forces students to distinguish between what they claim, what data supports the claim, and the mathematical principle that links the two. Using it for peer presentations builds academic language and models the kind of	Teacher records which students volunteer thumbs-down or thumbs-sideways and ensures each is cold-called or privately questioned. After consolidation, teacher poses an exit micro-task (1 minute, no notes): 'The midpoints of AA' and BB' are (2, 3) and (4, 5). Without drawing, what is the gradient of the mirror line? Write it down.' Students hold up mini-whiteboards or show on fingers. Teacher scans

Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	mirror line. The class uses a 'Claim–Evidence–Reasoning' (CER) frame printed on the board to evaluate each presentation: CLAIM — 'The equation of the mirror line is y = ___.' EVIDENCE — 'We know this because the midpoints of KK', LL', and MM' all lie on this line, at coordinates (__, __), (__, __), (__, __).' REASONING — 'This works because every point and its image are equidistant from the mirror line and the segment joining them is perpendicular to it.' After all three presentations, the teacher leads a 5-minute guided consolidation connecting back to the phenomenon: 'Let us go back to Lake Victoria. If AA' is the segment from a point on the boat to its mirror image in the water, the perpendicular bisector of AA' is exactly the waterline — the line of reflection. And now we can write that waterline as an equation: y = mx + c.'	WAIT TIME then ask for a 'thumbs agreement check' — thumbs up (agree with CER), thumbs sideways (not sure), thumbs down (disagree). Cold-call one 'thumbs sideways' student: 'What part of the reasoning are you unsure about?' During consolidation, use the document camera or board to draw the Lake Victoria cross-section: the boat above the axis, the image below, the waterline as y = mx + c. Ask: 'In our diagram, what does the value of m — the gradient of the mirror line — tell us about the direction of the waterline?' WAIT 15 seconds. Cold-call 2 students. Expected responses: 'It tells us the angle the waterline makes with the horizontal.' Affirm and extend: 'Exactly — a horizontal lake surface has gradient zero, so y = c. But on a tilted photograph, the apparent waterline could have any gradient.'	reasoning expected in congruence proofs in Lessons 6–8.	for correct answer (gradient = 1). Students who show incorrect answers are noted for Phase 5 support.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board displayed at the front of the room. The board retains sticky notes and annotations from Lessons 1–4. Each student receives two sticky notes of different colours. On the GREEN note: 'Write one thing you can now answer about our driving question that you could not answer before today.' On the ORANGE note: 'Write one new question today's lesson has raised for you.' Students place both notes in the Lesson 5 column of the DQB. The class then does a 3-minute 'gallery read' — students walk to the board, read the new notes silently, and place a small dot sticker next to any orange question that is also	Distribute sticky notes and give 3 minutes for individual writing — enforce silence so students do not copy neighbours. As students place notes, circulate and read over shoulders. During the gallery walk, stay near the board and listen to informal student conversations — note any persistent misconceptions (e.g., students confusing the perpendicular bisector with the mirror line itself). After reading the most-dotted orange questions aloud, say: 'Hold onto these questions — they are exactly what we will investigate in Lessons 6 and 7 when we move from finding mirror lines to using reflection to prove two figures are congruent.'	Colour-coded DQB Protocol: Using two distinct note colours for 'answers gained' vs. 'new questions raised' makes metacognitive progress visible and tangible. The dot-voting step democratises the process — students identify communally which open questions matter most — and mirrors the scientific practice of identifying gaps in current knowledge (NGSS SEP 1 — Asking Questions and Defining Problems).	Teacher reads all green notes after class to assess whether students correctly articulate the perpendicular-bisector–midpoint–equation chain. Green notes that only say 'I learned the mirror line' (without specifying the property) indicate surface-level understanding and flag students for targeted review at the start of Lesson 6. Teacher photographs the DQB after this phase for the ARES platform record.

	their question. The teacher identifies the two or three most-dotted orange questions and reads them aloud: 'These are the questions most of you still have — they will guide our next lessons.' The teacher explicitly updates the DQB with the key finding of Lesson 5 written in marker: 'Mirror line equation: $y = mx + c$. Found by constructing perpendicular bisectors of object-image segments.' The teacher also draws an arrow from this finding toward the driving question text: 'How does the mathematics of reflection explain the perfect mirror image on Lake Victoria?'	Write 'See Lessons 6 & 7' next to those orange notes on the DQB. This creates visible continuity in the inquiry storyline.		
Model Building Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their Cumulative Reflection Model — a large annotated diagram begun in Lesson 1 and updated each lesson. The model currently shows: the Lake Victoria shore scene, a labelled mirror line, arrows showing equal perpendicular distances from object to mirror and mirror to image, and the mapping notation $A \rightarrow A'$. Today, students add two new layers to their model: Layer A — Equation Layer: Students draw a coordinate grid alongside their shore diagram, plot one object-image pair from today's activity, construct the perpendicular bisector, and write the equation of the mirror line ($y = mx + c$) with m and c clearly labelled and explained in a legend. Layer B — 'Reverse Engineering' Annotation: Students add a label pointing to the mirror line that reads: 'If the waterline were hidden, I could find it by: (1) Joining each object point to its image. (2) Finding the midpoint of each segment. (3)	Distribute model folders and give students 2 minutes to silently re-read their existing models before adding new layers, saying: 'Read what past-you wrote. You are about to make your model smarter.' Circulate and prompt with specific questions: 'Where on your grid diagram would the waterline of Lake Victoria appear — above, below, or at $y = 0$? Justify.' WAIT 10 seconds after each prompt before accepting a response. For the extension question, direct early finishers to work independently without partner discussion first ('This is a thinking challenge — try it alone for 3 minutes before checking with your partner.'). In the final 3 minutes, ask 2 students to hold up their models and narrate one new annotation to the class. Close with the phenomenon connector: 'Next lesson, we will use everything in this model — the equal distances, the perpendicular property, the equation — to formally prove that the boat and its reflection are congruent figures.	Cumulative Model Revision (Progressive Modelling): Returning to and revising the same model across all 8 lessons embeds NGSS SEP 2 (Developing and Using Models). Each revision forces students to integrate new knowledge with prior knowledge rather than treating each lesson as isolated. The 'Reverse Engineering' annotation specifically targets procedural fluency with conceptual grounding — students must sequence the steps in their own words, which consolidates both the algorithm and its justification.	Teacher collects 6 model folders (a random stratified sample: 2 high, 2 mid, 2 low based on Phase 3 exit task) and checks two criteria before Lesson 6: (1) Is the equation of the mirror line written correctly with m and c identified? (2) Are the five 'reverse engineering' steps written in the student's own words and in the correct logical order? Folders with missing or misordered steps receive a written prompt returned at the start of Lesson 6.

	Drawing the perpendicular bisector. (4) Using two midpoints to calculate m. (5) Substituting to find c.' Students who finish both layers attempt the extension: 'Could two different mirror lines produce the same image from the same object? Explain your reasoning using the equal-distance property.' Models are kept in individual model folders for use in Lessons 6–8.	We are moving from describing the mirror image to proving it is mathematically identical.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your ARES teacher journal: (1) PHENOMENON CONNECTION — Did students spontaneously use the Lake Victoria waterline analogy when explaining the perpendicular bisector, or did they revert to abstract language? If reversion occurred, how might you re-anchor the context more frequently in Lesson 6? (2) PROCEDURAL VS. CONCEPTUAL — During the hands-on activity, did students construct perpendicular bisectors correctly but struggle to articulate WHY the bisector is the mirror line? If so, plan a 5-minute 'equal distance check' at the start of Lesson 6 where students measure both sides of the bisector with a ruler to confirm equal distances. (3) EQUATION WRITING — Which errors in calculating m or c were most common? Tally the error types from Phase 3 exit tasks (wrong gradient formula, arithmetic errors, incorrect substitution) and prepare one targeted example for the Lesson 6 warm-up. (4) DQB QUALITY — Were the orange 'new questions' on the DQB substantive and forward-looking (pointing toward congruence proofs), or surface-level (e.g., 'I don't understand gradients')? Surface-level questions signal a need to revisit linear equation skills before Lesson 6. (5) DIFFERENTIATION — Did the extension question ('Can two mirror lines produce the same image?') stretch high-attaining students productively? If most students did not reach it, consider making it a whole-class discussion opener in Lesson 6 rather than an individual extension.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed (via video and hands-on grid activity) that the perpendicular bisectors of the segments joining each object point to its corresponding image point all intersect at — or lie along — a single straight line. This line is the mirror line. Students measured and confirmed that each object point and its image are equidistant from this line.
What did I learn?	Students learned that the line of reflection can be determined from an object-image pair alone, without being told where the mirror is. The mirror line is the perpendicular bisector of the segment joining any object point to its image. Its equation can be written in the form $y = mx + c$, where m is found from the gradient of the line through two midpoints and c is found by substitution.
How does this explain the phenomenon?	Students explained how the equal-perpendicular-distance property of reflection makes it possible to 'reverse-engineer' the hidden waterline of Lake Victoria: by joining the boat's key points to their mirror images, finding the midpoints of those segments, and constructing perpendicular bisectors, the exact position and equation of the water surface can be determined mathematically — even when it is obscured by mist.

LESSON 6 (80 minutes): Same Shape, Same Size: Introducing Congruence	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students establish that reflection always produces a congruent image, and use formal congruence conditions (SSS, SAS, ASA, AAS, RHS) to prove that two triangles are identical in shape and size.
Knowledge	Students know that: (1) each point in the original figure is the same perpendicular distance from the line of reflection as its corresponding point in the image; (2) the image produced by reflection is congruent to the original figure; (3) the five congruence conditions (SSS, SAS, ASA, AAS, RHS) are sufficient to prove two triangles are congruent.
Skills	Students are able to: (1) measure sides and angles of triangles cut from card and sort them by congruence; (2) identify which congruence condition (SSS, SAS, ASA, AAS, or RHS) applies to a given pair of triangles; (3) write a formal congruence statement using the correct notation; (4) link the congruence of a reflected image to its original using perpendicular distance from the line of reflection.
Attitudes	Students appreciate the precision and economy of formal mathematical proof; they value collaborative investigation and respect differing sorting decisions made by peers before discussion.
Key Inquiry Question	How do you apply reflection to real-life situations, and what does it mean mathematically for two shapes to be exactly the same?
Purpose in Storyline	In lessons 1–5 students established HOW reflection works (mapping rules, perpendicular distance, the line y = mx + c). This lesson answers WHY the reflected image always looks like a perfect copy: the image is CONGRUENT to the original. Students now have the formal tools to PROVE sameness — completing the bridge between the intuitive mirror-image puzzle at Lake Victoria and rigorous geometric proof.
Safety Notes	Scissors are used to cut card triangles. Remind students to carry scissors with blades pointing downward and to cut away from the body. Ensure desks are clear of bags before distributing materials.

B. LESSON OVERVIEW

Lesson 6 opens by replaying the anchor photograph of the mirror-still Lake Victoria shoreline and asking students to recall their best current answer to the driving question. A short congruence video (approx. 5 min) then introduces the vocabulary of congruence in context. The core activity uses pre-cut card triangles labelled A–L (six congruent pairs, secretly arranged) that students measure with rulers and protractors, sorting them into "definitely the same" and "possibly the same" piles — mimicking the real-world challenge of a Kenyan roof-truss engineer checking whether factory-cut timber triangles are truly identical before installation. Through guided whole-class discussion the teacher formalises SSS, SAS, ASA, AAS, and RHS conditions. The lesson closes by returning to the phenomenon: students prove, using SSS, that the reflected boat triangle IS congruent to the original, because reflection preserves perpendicular distances and therefore all three side lengths.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Properties of](#)	Students are shown the anchor photograph of the Lake Victoria shoreline at dawn (Kisumu,	1. Project the Lake Victoria photograph. Say: 'Before we touch anything today, look at this image	Prediction-first activation (Elicit-before-Instruct): By requiring a written prediction before any	Observable indicator: All students have written at least two sentences in their books within 3

Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence: Justifying Algebraic Steps (PLIX) Source: CK-12 🔍 Search ARES for similar readings	projected on screen). Teacher uncovers the driving question board entry from Lesson 1: 'What rule does the water follow to make the reflection the same size?' Students individually write a 2-sentence prediction in their exercise books answering: 'If I measured every side and every angle of the reflected boat image and compared them to the real boat — what would I find? Why?' After 3 minutes of silent writing, students share predictions with a shoulder partner for 90 seconds. Three cold-call responses are taken and recorded on the board under two columns: 'Same measurements' / 'Different measurements'. Teacher does NOT confirm or correct — predictions stay visible for the full lesson.	we have been studying since Lesson 1. You now know more than you did then — use that knowledge.' 2. Read aloud: 'If I measured every side and every angle of the reflected boat image and compared them to the real boat — what would I find? Why?' 3. Allow 3 minutes of silent individual writing. Do NOT circulate — give thinking space. 4. Say: 'Share your prediction with your partner. You have 90 seconds. Go.' 5. After 90 seconds: 'I am going to cold-call three people. I want your prediction, not the right answer — your genuine thinking right now.' Cold-call three students by name (use class register — choose students who have not spoken in recent lessons). Record their exact words on the board. 6. Wait 10–12 seconds after each student speaks before moving on. 7. Say: 'These predictions will stay on the board. By the end of today, you will know exactly which column is correct — and you will be able to PROVE it.'	instruction, the teacher surfaces prior knowledge and misconceptions. Recording predictions publicly creates cognitive commitment — students are motivated to resolve the tension between their prediction and the evidence they gather. This directly feeds the inquiry cycle driving the 8-lesson sequence.	minutes. Teacher scans books during partner share (circulate briefly). Flag any student who writes 'I don't know' — prompt with: 'Think back to Lesson 3 — what happened to the distance from the line of reflection when we moved a point?' If the majority of cold-call responses suggest 'measurements will be the same', note this as a strong prior-knowledge base to build on. If responses are split or uncertain, note this as evidence that the formal congruence vocabulary is needed urgently.
Observe Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence:	PART A — Video (5 min): Students watch a short teacher-curated video introducing congruence, showing real-world examples including roof trusses on Kenyan market buildings, identical paving tiles at a Nairobi shopping mall, and triangular gusset plates on the Likoni Ferry ramp. Students are given a two-column observation note-frame ('What I notice' / 'What I wonder') and must record at least two entries in each column during the video.	PART A: 1. Say: 'You are about to watch a 5-minute video. Your job is NOT to sit back — your job is to be a detective. Use your note-frame. I want at least two entries in each column before the video ends.' Distribute note-frames. 2. Play video. Do not pause or comment during playback. 3. After video: 'Turn to your partner. Share one thing you noticed and one thing you wonder. 60 seconds.' Then cold-call two	Data-driven sorting (Structured Inquiry with Recording Scaffold): The recording table forces systematic measurement rather than visual guessing, mirroring scientific data collection. The Kisumu factory context makes the stakes concrete and Kenya-relevant, connecting abstract geometry to real structural engineering decisions. The deliberate inclusion of 'possibly the same' as a category honours genuine uncertainty and prevents false confidence — setting up the	Observable indicators: (1) Every student's recording sheet has all 72 measurement cells attempted (12 triangles × 6 values) by end of the activity — teacher spot-checks 3–4 sheets per group. (2) Each group has a written justification for each pair, not just labels. (3) Listen for groups debating 'close enough' versus 'exactly the same' — this is the productive tension that Phase 3 resolves. Flag any group that sorts purely by visual appearance without measuring — redirect: 'A factory machine cannot eyeball —

Justifying Algebraic Steps (PLIX) Source: CK-12 🔍 Search ARES for similar readings	PART B — Card Triangle Sorting Investigation (20 min): Each group of 4 students receives an envelope containing 12 pre-cut card triangles labelled A–L (secretly, these are 6 congruent pairs). Groups also receive a ruler, a protractor, and a recording sheet with a table (columns: Triangle label | Side 1 (cm) | Side 2 (cm) | Side 3 (cm) | Angle 1 (°) | Angle 2 (°) | Angle 3 (°)). Students measure every side and angle of every triangle and record results. They then sort triangles into groups they believe are 'definitely the same shape and size' and justify their grouping in writing. Context is framed as: 'You are a quality-control engineer for a Kisumu roof-truss factory. These triangles were cut by two different machines. Your job is to find out which triangles are truly identical before they go to the building site — a wrong match could cause a roof to collapse.'	pairs — one 'notice', one 'wonder'. PART B: 1. Distribute envelopes and recording sheets. Say: 'You are quality-control engineers at a roof-truss factory in Kisumu. These 12 triangles were cut by two machines. Some are identical — some might not be. Your task: measure carefully, record in your table, then sort them. Write your reason for every match you make.' 2. Circulate every 3–4 minutes. Use targeted prompts — do NOT give answers: - 'You have matched B and G — what measurement gave you the most confidence?' - 'Is measuring all six values always necessary? Could you get away with fewer? How few?' - 'What would happen on a building site if you matched two triangles that look the same but are not?' 3. With 5 minutes remaining: 'Pause measuring. Write your final groups on the board section assigned to your group.' 4. Allow all groups to post before any discussion.	need for formal congruence conditions in Phase 3.	what does your data say?'
Explain Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence:	PART A — Gallery Walk & Reveal (8 min): Groups circulate to view other groups' sorting boards (posted in Phase 2). Each student places a sticky dot next to any grouping they agree with and a question-mark sticker next to any they dispute. Students return to seats. Teacher reveals the correct pairs (A–G, B–H, C–I, D–J, E–K, F–L) and the class reconciles agreements and disputes.	PART A: 1. Say: 'You have 4 minutes to walk and place your dots and question marks. Move silently and read carefully.' Set a timer. 2. After return: cold-call two students on disputed groupings. Ask: 'What measurement made you uncertain?' Wait 12 seconds after each response. 3. Reveal correct pairs. Say: 'Those of you who were uncertain	Claim–Evidence–Reasoning (CER) framework applied through the gallery walk dispute process, and Worked Example with Student Attempt (for congruence condition uniqueness proof). The gallery walk makes student thinking public and falsifiable — a scientific norm. The uniqueness-attempt strategy (can you draw a DIFFERENT triangle?) is a constructivist proof-by-contradiction scaffold that makes formal conditions feel	Observable indicators: (1) During gallery walk, every student places at least one dot and one question-mark sticker — passive non-engagement is visible and addressable. (2) During SSA counter-example, students should be able to produce a second, non-congruent triangle — if no one can, teacher draws two possible triangles satisfying SSA to show it fails. (3) In Part C, the written congruence statement must

Justifying Algebraic Steps (PLIX) Source: CK-12 🔍 Search ARES for similar readings	PART B — Formalising Congruence Conditions (15 min): Teacher leads a guided whole-class investigation. Using a projected Geogebra diagram (or hand-drawn board diagram), teacher demonstrates each congruence condition using triangles from the card set: SSS (three sides), SAS (two sides and included angle), ASA (two angles and included side), AAS (two angles and non-included side), RHS (right angle, hypotenuse, one side). For each condition, teacher draws the triangle, labels it, and asks students to attempt to draw a DIFFERENT triangle satisfying the same condition — proving uniqueness (or lack thereof, e.g., SSA is not a valid condition). PART C — Phenomenon Link (7 min): Teacher returns to the Lake Victoria photograph and draws a simple triangle on the 'boat' above the waterline (vertices P, Q, R) and its reflected image (P', Q', R') below. Students use their rulers on a printed diagram to confirm: PP', QQ', RR' are all bisected perpendicularly by the waterline. Teacher asks: 'What do you now know about the sides of triangle PQR and triangle P'Q'R'?' Students write a formal congruence statement and identify which condition proves it.	— your uncertainty was mathematically valid. Today we are going to find out exactly how MANY measurements you need to be 100% certain.' PART B: 1. Write on board: 'CONGRUENCE CONDITION — a minimum set of measurements that GUARANTEE two triangles are identical.' 2. For each condition (SSS → SAS → ASA → AAS → RHS): a. Draw and label the triangle. b. Ask: 'Can anyone draw a DIFFERENT triangle with these same measurements?' Give 10 seconds wait time. Cold-call one student to attempt it on the board. c. Confirm or address. Say: 'This is why SSS [or the relevant condition] is a PROOF — not a guess.' 3. After all five conditions: 'Write all five conditions in your notes with one example each — you have 3 minutes.' PART C: 1. Project Lake Victoria diagram with triangle PQR and image P'Q'R'. 2. Say: 'We established in Lesson 3 that every point and its image are the same perpendicular distance from the line of reflection. What does that tell us about the lengths PQ and P'Q'? QR and Q'R'? PR and P'R'?' Wait 15 seconds. Cold-call three students (one per side pair). 3. Say: 'Write the formal congruence statement in your book and state which condition	necessary rather than arbitrary. The phenomenon link in Part C closes the loop: students do not just learn SSS — they USE it to prove the very puzzle that opened the unit.	include the ≅ symbol, matching vertex order, and named condition — teacher checks 5 books during the 2-minute writing time. A student writing 'PQR = P'Q'R'' (using = instead of ≅) is a specific, correctable misconception to address immediately.

		you are using.' Allow 2 minutes. Cold-call one student to read their statement. Write the model answer on the board: 'Triangle PQR ≅ Triangle P'Q'R' by SSS.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence: Justifying Algebraic Steps (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board displayed at the front of the room. Each student receives one sticky note. They are asked to write a response to the prompt: 'What NEW evidence from today's lesson helps you answer the driving question — and what question do you still have?' Students post their notes under the two standing columns: 'Evidence we now have' and 'Questions we still have'. The class does a 3-minute read-around of new notes. Teacher highlights two or three notes that directly name 'congruence' or a condition (SSS, SAS, etc.) and reads them aloud, connecting them explicitly back to the Lake Victoria phenomenon. Teacher also circles any new questions that will be addressed in Lessons 7–8 (e.g., 'Does congruence work for non-triangles?' 'Can we prove congruence without measuring everything?').	1. Hand each student a sticky note. Say: 'Two things on your note: one piece of NEW evidence you gathered today that helps answer our driving question, and one question you still have. You have 2 minutes. Go.' 2. Students post notes. Allow 1 minute for silent read-around (students walk to board, read, return). 3. Teacher reads 2–3 selected notes aloud. For each: 'This student wrote [quote]. How does this connect to the boats on Lake Victoria?' Wait 10 seconds. Take one volunteer response. 4. Point to the 'Questions we still have' column. Say: 'These questions — especially [name a specific new question] — are exactly what Lessons 7 and 8 are built to answer. Your curiosity is driving this course.' 5. Do NOT remove any sticky notes — the cumulative board is a learning artefact.	Driving Question Board as a living knowledge organiser: The DQB makes the growth of collective understanding visible over time. By requiring both 'evidence' and 'remaining questions', the strategy models scientific practice — knowledge is always provisional and cumulative. Publicly honouring student-generated questions reinforces that inquiry is not teacher-driven but learner-driven.	Observable indicator: Every student posts a note with both components (evidence + question). Teacher scans for notes that only restate a procedure ('I learned SSS') without connecting to the phenomenon — these students need the Lesson 7 warm-up to revisit the link. Notes that ask rich structural questions ('Does the line of reflection have to be straight for congruence to hold?') indicate readiness for extension in Lesson 8.
Model Building Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of	Students open their cumulative model booklet (started in Lesson 1). On a new page headed 'Lesson 6 — Congruence', they complete three tasks: (1) DIAGRAM — Draw triangle PQR above a horizontal line of reflection (the waterline) and its image P'Q'R' below. Label corresponding sides and mark equal lengths with tick marks. Write the congruence statement: Triangle PQR ≅ Triangle P'Q'R'.	1. Say: 'Open your model booklets to a new page. Title it Lesson 6 — Congruence. You have three tasks — they are written on the board. Read them before you start.' Read all three tasks aloud. 2. Allow 10 minutes of independent work. Circulate and check: - Diagram: corresponding vertices must be labelled with matching letters (P↔P', Q↔Q', R↔R'). Redirect if students label	Cumulative Model Revision (Progressive Model Building): The model booklet is the student's running scientific explanation — analogous to a researcher's lab notebook. Requiring a 'revision caption' each lesson forces metacognitive articulation: students must name what changed in their understanding, not just what they did. The tick-mark convention for equal sides is a formal mathematical notation habit	Observable indicators: (1) Diagram shows correct perpendicular positioning of image below waterline (not a random placement). (2) All three pairs of corresponding sides are marked with matching tick marks. (3) Congruence statement uses ≅ (not =) and correct vertex order. (4) Revision caption explicitly names at least one congruence condition by name. Teacher flags booklets where the diagram is

[Equality and Congruence: Justifying Algebraic Steps (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	(2) CONDITIONS TABLE — Draw a 2-column table listing all five congruence conditions (SSS, SAS, ASA, AAS, RHS) with a one-sentence explanation of each in their own words. (3) MODEL REVISION CAPTION — Below their diagram, write 2–3 sentences updating their model: 'In Lessons 1–5 I showed HOW reflection maps points. Now I can add that the image is always CONGRUENT to the original because reflection preserves all distances, so all side lengths stay the same — which means SSS is satisfied.' The teacher collects model booklets at the end and returns them with a written stamp/tick confirming the diagram is correct — booklets are returned at the start of Lesson 7.	arbitrarily. - Tick marks: both sides of each corresponding pair must carry the same number of ticks. - Congruence statement: vertex ORDER must match. Flag 'Triangle PQR ≅ Triangle Q'P'R'' as incorrect — order matters. 3. With 2 minutes remaining: 'Write your revision caption. Start with the sentence stem on the board: In Lessons 1–5 I showed HOW reflection maps points. Now I can add…' 4. Collect booklets. Say: 'Your model is growing every lesson — by Lesson 8 you will have a complete mathematical explanation of the Lake Victoria phenomenon, and you will have proved it.'	built progressively across lessons.	correct but the caption is generic ('I learned about congruence') — these students can state facts but cannot yet narrate conceptual change, which is the target of Lesson 7's opening activity.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes before Lesson 7: \n\n1. PREDICTION QUALITY: Were students' Predict-phase responses grounded in Lessons 1–5 evidence (e.g., 'perpendicular distance is preserved, so lengths are preserved'), or were they still intuitive ('the reflection looks the same')? If mostly intuitive, plan a 5-minute Lesson 7 warm-up that explicitly bridges perpendicular distance → preserved length → SSS. \n\n2. SORTING ACCURACY: How many groups correctly identified all six congruent pairs from measurement alone? If fewer than half the groups succeeded, this suggests measurement skills (ruler and protractor use) need reinforcement — consider a short protractor-accuracy drill in Lesson 7. \n\n3. CONDITION CONFUSION: Did students consistently confuse ASA and AAS, or mistake SSA for a valid condition? If yes, prepare a targeted counter-example card for Lesson 7's warm-up: two non-congruent triangles satisfying SSA that students must distinguish. \n\n4. PHENOMENON LINK: Could every student write the congruence statement 'Triangle PQR ≅ Triangle P'Q'R' by SSS' independently by the end of Phase 3? If not, identify which students could not and pair them deliberately in Lesson 7's group work. \n\n5. MODEL BOOKLETS: How many booklets show a correct diagram with matching tick marks AND a substantive revision caption? This is your clearest evidence of deep understanding versus surface procedural compliance. Use the proportion as a gauge for how much Lesson 7 needs to consolidate before advancing to congruence proofs with formal two-column notation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed (via the anchor photograph, the congruence video, and the card-triangle sorting investigation) that triangles which match on all three sides — as occurs between a reflected image and its original across the Lake Victoria waterline — are always identical in shape and size, regardless of how the triangles are oriented.
What did I learn?	Students learned that reflection always produces an image that is congruent to the original figure, because every point and its

	image are the same perpendicular distance from the line of reflection, preserving all side lengths. They learned the five formal congruence conditions (SSS, SAS, ASA, AAS, RHS) that are sufficient to prove two triangles identical, and that SSS alone is enough to prove the reflected boat image congruent to the original boat.
How does this explain the phenomenon?	Students explained that the reflected boat image on Lake Victoria is a perfect, same-sized copy of the original because reflection is a distance-preserving transformation: each vertex of the boat maps to a vertex in the image at an equal perpendicular distance from the waterline, so all three side lengths of the image triangle equal all three side lengths of the original triangle — satisfying SSS — which proves the two triangles are congruent (Triangle PQR ≅ Triangle P'Q'R').

LESSON 7 (40 minutes): Proving It: Applying Congruence Conditions	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to select and apply the correct congruence condition to prove that two triangles are congruent, and to connect this formal proof to the reason why a reflected image is always an exact copy of the original.
Knowledge	Learners will identify which of the five congruence conditions (SSS, SAS, ASA, AAS, RHS) applies to a given pair of triangles, state formal congruence statements using correct notation, and explain why reflection is a rigid transformation that guarantees congruence.
Skills	Learners will analyse diagrams to extract known sides and angles, construct step-by-step congruence proofs with justification, and critique and defend proofs presented by peers.
Attitudes	Learners will appreciate that mathematical proof requires precision and logical reasoning, and will develop confidence in presenting and defending a mathematical argument.
Key Inquiry Question	How do you apply reflection to real-life situations? — addressed by formalising why the Lake Victoria boat reflection is always congruent to the original boat, using congruence conditions as proof.
Purpose in Storyline	This lesson is the proof-writing climax of the sequence. Students have observed the phenomenon, discovered reflection rules, and learned congruence conditions. Now they apply those conditions in structured proofs and peer argumentation, producing the mathematical explanation for why the reflected boat image is always a perfect, same-sized copy. This evidence is added to the Driving Question Board, nearly completing the answer to the driving question.
Safety Notes	No significant safety concerns. Ensure rulers and compasses are handled responsibly during any diagram construction. Classroom movement during group rotations should be managed calmly.

B. LESSON OVERVIEW

Learners open by revisiting the Lake Victoria photograph and recalling the five congruence conditions from Lesson 6. In the Predict Phase, they examine three pairs of triangles drawn on the board and predict which congruence condition — if any — proves each pair congruent, recording reasons before any calculation. In the Observe Phase, groups work through a structured proof worksheet featuring contexts drawn from Kenyan life (a roof-truss over a Kisumu market, a symmetrical Maasai shield, and a reflected fishing-boat outline on a coordinate grid), annotating diagrams, marking equal sides and angles, and writing formal congruence statements. A peer argumentation task then requires groups to examine another group's proof, identify any errors in condition selection or notation, and present a challenge or confirmation. In the Explain Phase, the class constructs a shared proof that the Lake Victoria boat reflection is always congruent to the original, identifying SSS as the decisive condition and formalising the statement. The Driving Question Board is updated with this formal proof. In the Model Building Phase, learners add a proof template to their cumulative Cartesian models, linking every reflection rule learned since Lesson 2 to the guarantee of congruence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	The teacher projects the Lake Victoria dawn photograph used in	Display the three triangle pairs on the board BEFORE distributing	Predict-before-observe routine activates prior knowledge of	Teacher scans notebooks during the 3-minute silent prediction to

📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Equality and Congruence: Justifying Algebraic Steps (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Lesson 1 and asks: 'We said the reflection is always the same size. Today we prove it — using the congruence conditions you learned last lesson.' Three pairs of triangles are drawn on the board (or projected on a chart): Pair A has three side lengths marked equal on each triangle (SSS situation); Pair B has two sides and an included angle marked (SAS situation); Pair C has a right angle, the hypotenuse, and one side marked (RHS situation). Learners work individually for 3 minutes, writing in their notebooks: (1) which condition they think applies to each pair, (2) the letters of the corresponding vertices in the correct order, and (3) whether they are certain or uncertain. They then share predictions with a shoulder partner before the teacher takes a quick show-of-hands census for each pair.	any worksheet so attention is focused. Say: 'Do not measure anything yet — reason only from the marks already on the diagram. Write your prediction and your reason.' WAIT 3 full minutes in silence, circulating to observe notebook entries without commenting. After sharing with partners, ask: 'Who changed their mind after talking with their partner? What made you change?' WAIT 20 seconds. Probe with: 'For Pair C — what is special about the right angle and hypotenuse that makes RHS a separate condition from SAS?' WAIT 30 seconds. Record a tally of class predictions on the board to revisit after the Observe Phase.	congruence conditions from Lesson 6. The show-of-hands census makes class thinking visible and creates productive uncertainty where the class is split, motivating careful observation.	identify students who are confusing condition names, writing vertices in incorrect correspondence order, or who are uncertain about RHS. These students are noted for targeted support during the Observe Phase group work.
Observe Phase 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Equality and Congruence: Justifying Algebraic Steps (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Groups of three receive the structured proof worksheet. It contains four tasks. Task 1 (Kisumu Market Roof Truss): A diagram shows a symmetrical triangular roof truss — two triangles sharing a common vertical strut. Side lengths are labelled in metres. Learners identify three pairs of equal sides and write: 'Triangle ABC is congruent to Triangle XYZ by SSS because AB = XY = 4 m, BC = YZ = 6 m, and AC = XZ = 7 m.' Task 2 (Maasai Shield): A symmetrical shield outline is drawn on a grid. The line of symmetry is marked. Two triangular sections either side of the axis are given with two sides and the included angle. Learners write the SAS proof. Task 3 (Reflected Fishing Boat on Grid): A	Distribute worksheets and say: 'For every proof you write, you need three things: the condition name, the pairs of equal elements with their values, and the formal statement. Missing any one of these means the proof is incomplete.' Circulate during Task 1 — listen for groups who name the condition but fail to list all three pairs of equal sides. Prompt: 'You have said SSS — but where are the three S-es in your written proof?' WAIT 15 seconds. During Task 3, prompt any group struggling with the distance formula: 'Can you count the horizontal and vertical distances between the points on the grid first? Then use Pythagoras.' During the peer argumentation rotation, say: 'You are not rewriting	Scaffolded progression from a pure-geometry context (roof truss) to a coordinate-geometry context (reflected boat) connects abstract proof writing to the anchoring phenomenon. The peer argumentation task develops mathematical reasoning and communication by requiring learners to evaluate reasoning rather than just produce it.	Collect the Task 4 half-sheets with peer comments after the rotation. These reveal whether students can identify errors in condition selection, notation, and correspondence — key indicators of depth of understanding. Teacher notes recurring errors for the Explain Phase discussion.

	coordinate grid shows triangle PQR (the boat outline) with vertices at P(2,3), Q(5,3), R(5,7) and its reflection P'(-2,3), Q'(-5,3), R'(-5,7) across the y-axis. Learners calculate all three side lengths for both triangles using the distance formula (or counting on the grid), confirm they are equal, and write the SSS congruence statement. Task 4 (Peer Argumentation): Each group writes their proof for Task 2 on a half-sheet of paper. Groups rotate papers clockwise. The receiving group checks: Is the condition named correctly? Are corresponding vertices in the right order? Is each equal element explicitly stated? They write one 'Challenge or Confirm' sentence and return the paper. Groups then have 2 minutes to respond to feedback.	their proof — you are a mathematics examiner. Write exactly one sentence that either confirms it is correct or identifies the first error you find.' After rotations, bring the class together: 'Did any group find an error? What was the most common mistake you saw?' WAIT 30 seconds for hands. Affirm correct challenges and ask the original group to correct their proof aloud.		
Explain Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence: Justifying Algebraic Steps (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. The teacher projects Task 3 (the reflected fishing boat on the coordinate grid) on the board. A volunteer group presents their SSS proof step by step. The teacher then asks the whole class: 'This is not just a proof about these three points. What does it tell us about EVERY boat on Lake Victoria — no matter how complicated its shape?' Learners discuss in pairs for 90 seconds, then the teacher takes responses. The class co-constructs a general explanation: when a shape is reflected across any straight line, every point and its image are the same distance from the mirror line. Therefore every side length is preserved. If all three sides of a triangle in the original equal all three sides of the corresponding	After the volunteer group presents, resist correcting immediately. Ask the class: 'Does anyone want to add to or challenge what was just presented?' WAIT 20 seconds. If no challenge arises and the proof is correct, validate and extend: 'This proof works for these three specific points. Help me generalise — why does this guarantee congruence for the entire boat outline, not just one triangle?' WAIT 30 seconds. If students struggle, scaffold: 'The boat outline is made of many triangles. If SSS works for every triangle, what follows?' After the boxed summary is written, return to the phenomenon photograph and say: 'Point to the mirror line. Point to one original vertex and its image. How far is each from the waterline? Who can say in one	The move from a specific coordinate proof to a general claim about all reflections builds scientific-style generalisation. Returning to the phenomenon photograph at the end of the explanation closes the loop from concrete observation (Lesson 1) to formal proof (Lesson 7), giving students a clear sense of explanatory progress.	Teacher listens during the pair discussion for the language students use: do they say 'it is the same size' (informal) or 'all side lengths are preserved, so SSS applies' (formal)? This distinguishes surface familiarity from deep understanding. The boxed summary in notebooks is checked during the DQB phase.

	triangle in the image, SSS guarantees congruence. The teacher writes on the board: 'Reflection is a rigid transformation. Rigid means distances do not change. Therefore: original triangle is ALWAYS congruent to its reflected image (SSS).' Students copy this into their notebooks as a boxed summary statement. The teacher then returns to the board tally from the Predict Phase: 'Were your predictions correct? What surprised you about Pair C — and why is RHS not just a special case of SAS?'	sentence why the image is always the same size?' WAIT 20 seconds. Accept and refine student language toward: 'Reflection preserves distance, so all sides stay equal, so SSS holds, so the image is always congruent.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence: Justifying Algebraic Steps (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The DQB — maintained since Lesson 1 — is displayed at the front. Students individually read the current questions and sticky notes. The teacher reads aloud three questions that appeared early in the sequence: (1) 'Why does the reflection always look exactly the same size?' (2) 'What does it mean mathematically for two shapes to be exactly the same?' (3) 'Could you prove that the boat and its image are identical?' Learners write on new sticky notes the evidence from today that answers each question, using the formal language from the Explain Phase. Selected students place their notes on the DQB beside the original questions. The teacher asks: 'Look at the board. Which of our opening questions from Lesson 1 can we now answer fully — with a mathematical proof?' Learners identify questions 1, 2, and 3 as now answered. The teacher points to any remaining open questions: 'What is still left for Lesson 8?' Learners note that applications to real-world design	Stand beside the DQB and say: 'This board has been collecting our thinking since Lesson 1. Today we get to answer three of the biggest questions — but only if we use the right language. Look at the boxed summary in your notebook. Use those exact words on your sticky note.' WAIT 2 minutes for writing. As students post notes, read two aloud and ask the class: 'Is this answer precise enough to count as a mathematical proof, or is it still informal?' WAIT 20 seconds. Prompt precision: 'If someone said the image is the same size because water is flat — is that a mathematical explanation? What is missing?' Guide students toward: a proof requires naming the condition (SSS) and listing the equal elements. End by pointing to Lesson 8 preview: 'Tomorrow we use congruence to find distances we cannot measure directly — like the width of a river. How might that work?'	The DQB update makes the explanatory progress of the entire unit visible in one place, reinforcing the coherence of the storyline. Distinguishing informal answers from mathematical proofs on the DQB builds metacognitive awareness of what counts as sufficient explanation in mathematics.	Sticky notes are a written formative artifact. The teacher collects and reads those not posted to the board, checking whether students can write a concise, formally correct explanation that references a specific congruence condition. Sticky notes that are purely informal are flagged for follow-up at the start of Lesson 8.

	and inaccessible distance problems are still to be explored.			
Model Building Phase 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Equality and Congruence: Justifying Algebraic Steps (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	Students open their cumulative Cartesian model — the A3 grid sheet built up across Lessons 2 through 6, which currently shows the boat-shape reflected across multiple mirror lines with all transformation rules annotated. Today they add a Proof Panel in the bottom-right corner of the sheet. The Proof Panel contains: (a) a miniature version of the Task 3 fishing-boat grid, (b) the three side-length calculations confirming SSS, (c) the formal congruence statement written in correct notation, and (d) the boxed general claim: 'Reflection is a rigid transformation. All side lengths are preserved. Therefore original figure is congruent to reflected image (SSS).' Below this, learners write a one-sentence connection: 'This is why the Lake Victoria boat and its mirror image are always the same size — reflection preserves every side length, so SSS is always satisfied.' Learners who finish early are invited to add a second Proof Panel using SAS for an isosceles triangle reflected across its axis of symmetry.	Distribute the cumulative model sheets (or direct students to their folders) and say: 'Your model has shown how to find the image of every point. The Proof Panel is the final piece — it explains WHY the image is always the same as the original.' Circulate and check that the formal congruence statement uses the triangle congruence symbol and lists vertices in correct correspondence order. If a student writes the symbol incorrectly, ask: 'Read your statement aloud. Does the order of vertices in Triangle PQR match the order in Triangle P-prime Q-prime R-prime? Check your corresponding sides.' WAIT 15 seconds. In the final 2 minutes, ask two students to read their one-sentence connection to the class. Say: 'Keep this model. In Lesson 8 you will add one final panel showing a real-world application. By the end of next lesson, your model will be a complete mathematical explanation of the Lake Victoria phenomenon.'	The cumulative model is a boundary object that integrates every lesson into a single coherent artefact. Adding the Proof Panel makes the logical relationship between reflection rules and congruence proofs spatially explicit — students can literally point from a reflected point on the grid to the proof that explains why the image is congruent.	Teacher photographs or collects the updated model sheets at the end of the lesson. The Proof Panel is assessed against three criteria: (1) correct side-length calculations, (2) correct congruence condition named, (3) formal congruence statement with vertices in correct correspondence. Results inform grouping and differentiation for Lesson 8 application tasks.

D. TEACHER REFLECTION

After the lesson, consider: Did learners distinguish between naming a congruence condition and writing a complete formal proof with all equal elements listed? Were students able to write vertex correspondence correctly in the congruence statement, or did they name the triangles in random order? Did the peer argumentation task surface genuine mathematical disagreement, or did groups simply confirm each other? Did the connection between SSS and the Lake Victoria phenomenon feel mathematically satisfying to students, or did it still feel like a claim rather than a proof? For Lesson 8, identify which students still rely on informal language ('same size') rather than formal proof language ('SSS because AB = XY, BC = YZ, AC = XZ') — these students need a brief revision prompt at the start of the applications lesson. Also consider whether the proof template on the cumulative model is clear enough to support independent application in Lesson 8 without re-teaching.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed three pairs of triangles and a reflected fishing-boat coordinate diagram, and we saw that the side lengths of the original triangle and its reflected image are always equal.
What did I learn?	We learned how to write a formal congruence proof by naming the correct condition (SSS, SAS, ASA, AAS, or RHS), listing all pairs of equal elements with their values, and writing the congruence statement with vertices in correct correspondence order.
How does this explain the phenomenon?	We explained that the Lake Victoria boat reflection is always a perfect, same-sized copy because reflection is a rigid transformation that preserves all side lengths — so SSS is always satisfied, and the reflected image is always congruent to the original boat.

LESSON 8 (80 minutes): Mathematics in the Real World: Applications of Reflection and Congruence	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To apply the mathematics of reflection and congruence to real-world contexts — including symmetrical design, inaccessible distance calculation, and verification of building plans — and to synthesise all unit learning into a final, complete explanation of the anchoring phenomenon.
Knowledge	Students know that: (1) each point in a figure and its reflected image are equidistant from the line of reflection, measured perpendicularly; (2) the line of reflection is expressed in the form $y = mx + c$; (3) the image produced by reflection is always congruent to the original figure — same shape, same size, but flipped; (4) congruence can be established and proved using the properties of reflection.
Skills	Students are able to: apply reflection rules to design symmetrical patterns; use congruent triangles to calculate distances that cannot be measured directly; verify whether two geometric figures are congruent; present a complete written and verbal explanation of why a reflected image is always congruent to the original; revise and finalise a cumulative unit model.
Attitudes	Students appreciate the power of mathematical reasoning as a tool for solving real-world problems in design, engineering, and environmental observation; they show confidence in using precise mathematical language to explain natural phenomena.
Key Inquiry Question	How do you apply reflection to real-life situations?
Purpose in Storyline	This is the final lesson of the unit. Students arrive having gathered evidence across seven lessons about the properties of reflection and congruence. Today they apply that evidence to three Kenya-relevant real-world tasks, complete the Driving Question Board by confirming which original questions have been fully answered, compare their Lesson 1 intuitive model with their final cumulative model, and write a rigorous, full explanation of the anchoring phenomenon using precise mathematical language.
Safety Notes	No physical hazards. Remind students to use rulers carefully when drawing construction lines. Ensure classroom furniture is arranged to allow safe group movement during the gallery-walk phase.

B. LESSON OVERVIEW
Lesson 8 is the capstone lesson of the Sub-Strand 1.4 Congruence unit. Students open by watching a short applications video that situates reflection and congruence in three Kenyan contexts: Maasai bead jewellery design (symmetry), surveying inaccessible distances across the Rift Valley (congruent triangles), and verifying a building blueprint in Nairobi (congruence properties). They then work in groups on three corresponding real-life problem tasks. A whole-class DQB Completion Ceremony follows, in which every original student question posted in Lesson 1 is revisited and formally answered using the mathematical language of the unit. Students then place their Lesson 1 intuitive boat-reflection model side-by-side with their final cumulative model and narrate the differences in a structured Model Comparison protocol. The lesson closes with an individual Exit Explanation Task — a full written account of the anchoring phenomenon ("The Perfect Mirror Image at Lake Victoria's Shore") — in which students must use the terms: line of reflection, perpendicular distance, image, congruence, and $y = mx + c$.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students are shown three	Display the three photographs.	Activation of Prior Knowledge —	Listen for: students spontaneously

📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cofunction Identities and Reflection (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	photographs projected on the board: (A) a Maasai woman wearing an elaborate beaded necklace from Kajiado County, (B) a land surveyor using equipment on the escarpment above the Rift Valley near Naivasha, (C) an architect's blueprint of a symmetrical building in Nairobi's Westlands district. Before any explanation, students write individual predictions in their notebooks, responding to the prompt: 'In what way do you think reflection or congruence appears in each of these three images? Use the mathematical language we have built this unit.' Students are given 3 minutes of silent writing time, then share predictions with a shoulder partner for 2 minutes.	Say exactly: 'Before we watch the video, I want you to predict — using the mathematical words we have been building all unit — where you think reflection or congruence is hiding in each of these three real-world images. You have three minutes of silent writing. Begin.' [Allow 3 minutes strict silence.] After writing: 'Share your prediction with your shoulder partner. You each have one minute.' [Allow 2 minutes.] Cold-call 3 students (one per photograph) to share predictions with the class. For each, ask: 'What specific mathematical property are you seeing? Can you name the line of reflection, even approximately?' [Allow 10 seconds WAIT TIME after each question before accepting answers.] Record key predicted ideas on the board in a column labelled 'Our Predictions — Before Video.'	By predicting before viewing, students surface what they have consolidated across Lessons 1–7 and commit to specific mathematical claims. This creates cognitive investment in the video and primes them to notice where their predictions are confirmed or refined.	using the phrases 'line of reflection,' 'perpendicular distance,' 'congruent,' and 'y = mx + c.' A student who can identify a line of symmetry in the bead pattern and describe it as 'a line of reflection where each bead is equidistant from the line' is demonstrating target-level understanding. Students who rely only on everyday language ('it looks the same on both sides') are flagged for targeted support during the group task phase.
Observe Phase 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cofunction Identities and Reflection (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Students watch a 6-minute applications video produced for this lesson. The video covers three segments: (1) A Maasai beadwork artisan in Kajiado explains how she ensures her necklaces are symmetrical, unknowingly describing the line of reflection; the video then overlays a coordinate grid on the beadwork and shows that corresponding beads are equidistant from the central line. (2) A surveyor at the Rift Valley escarpment demonstrates how congruent triangles are used to calculate the width of an inaccessible gorge without crossing it. (3) An architect in Nairobi explains how a building plan is checked for congruence between the left and right wings. After the video, students return to	Before playing the video, say: 'As you watch, keep your notebook open to a two-column table: left column — what you observe; right column — which mathematical property explains it. You will need this for the group tasks.' [Play the 6-minute video.] After the video, pause and say: 'Take 90 seconds — add any final observations to your table.' [Allow 90 seconds.] Direct students to the DQB: 'Walk slowly to the board and read each question card. Do not talk yet — just read and think.' [Allow 2 minutes quiet reading.] Return to seats. Distribute the DQB summary sheets. Say: 'Annotate every question. Write the lesson number where you found the evidence. You have 4 minutes.' [Allow 4 minutes.] Cold-call 4	Evidence Mapping — Students explicitly link each original driving question to specific lesson evidence. This metacognitive activity makes the cumulative logic of the storyline visible, helping students see that their understanding has been systematically built, not acquired all at once.	Collect DQB summary sheets at the end of the lesson. During the annotation phase, circulate and check: can students cite specific lesson evidence (e.g., 'Lesson 3 — we measured perpendicular distances from points to the line of reflection')? Students who write vague annotations ('we learned about reflection') need targeted questioning: 'Which specific activity gave you evidence? What did you measure or calculate?'

	the Driving Question Board (DQB) posted since Lesson 1 and, in silence, read every question card. They then individually annotate a printed DQB summary sheet: for each question, they write 'Answered — Evidence from Lesson ___' or 'Still unclear — because ___.'	students on different questions: 'Which lesson gave us the clearest evidence for that question? What was the evidence?' [10 seconds WAIT TIME per question.]		
Explain Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of four on three structured application tasks (groups rotate through all three, spending approximately 10 minutes per task). TASK A — Maasai Bead Pattern Design: Given a half-pattern of a Maasai necklace plotted on a coordinate grid, students reflect the pattern across the line y = x to complete the full symmetrical design. They must list the coordinates of five key points and their images, and verify that corresponding points are equidistant from the line y = x. TASK B — Inaccessible Distance Across the Gorge: Given a diagram of a gorge near Naivasha where two congruent triangles have been constructed by a surveyor, students use the congruence of the triangles to calculate the width of the gorge. They must state which congruence condition (SAS, ASA, SSS, or RHS) justifies their calculation and write one full justification sentence. TASK C — Nairobi Building Blueprint Verification: Given coordinate pairs for the left and right wings of a building, students verify whether the right wing is a true reflection of the left wing across the line x = 7. They must check perpendicular distances for all given vertices and write a formal conclusion sentence beginning: 'The right wing IS / IS	Arrange the room into three stations before the lesson. Assign groups. Say: 'Each station has a real-world problem from Kenya. Your group has 10 minutes at each station. You must produce written working — every calculation must be shown. At the end of each task, one group member must be ready to explain your group's answer to the class. Rotate when I signal.' [Set a visible 10-minute timer.] During rotation, circulate and use targeted prompts: At Task A — 'How are you finding the perpendicular distance from this bead-point to the line y = x? Show me the calculation.' At Task B — 'Which congruence condition are you using? Read me your justification sentence.' At Task C — 'What does x = 7 look like on the grid? Is it vertical or diagonal? How does that affect your perpendicular distance calculation?' After all rotations, cold-call one group per task to present their answer. [Allow 10 seconds WAIT TIME before any student speaks.] After each presentation, ask the class: 'Does this group's answer connect back to the phenomenon at Lake Victoria? How?'	Contextualised Problem-Solving with Structured Justification — By embedding mathematical procedures in three distinct Kenyan real-world contexts, students see that reflection and congruence are not abstract rules but tools for solving genuine problems. The requirement to write justification sentences forces students to move from procedural to explanatory thinking, which is the sensemaking goal of the final lesson.	Collect written work from all three stations. Indicators of target understanding: (1) Task A — correct coordinates listed for all five image points, with a written perpendicular-distance check showing equality for at least three points. (2) Task B — correct distance calculated, correct congruence condition named, and a grammatically complete justification sentence. (3) Task C — all vertices checked, correct conclusion written with mathematical reasoning. Students who complete calculations correctly but cannot write a justification sentence are identified for the exit task scaffolding support.

	NOT congruent to the left wing because…'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class gathers around the DQB (or it is projected on the board). The teacher reads each original question card aloud, one by one. For each question, a student volunteer (or cold-called student) provides the class answer in one or two sentences using precise mathematical language. The class votes with thumbs up/sideways/down: 'Fully answered / Partially answered / Still unclear.' Questions that receive a 'fully answered' consensus are marked with a green checkmark (or green sticky note). If any question receives 'still unclear' from more than one-third of the class, the teacher provides a concise final resolution. After all questions are addressed, the teacher asks: 'Which single mathematical idea was the most important key to unlocking the whole phenomenon?' Students write their answer in one sentence in their notebooks, then share with a partner before a whole-class cold-call of three responses. The five original puzzling questions from Lesson 1 are also formally resolved: (1) The reflection is always the same size because reflection preserves distances — it is an isometry. (2) The rule the water surface follows is: every point maps to a point equidistant from the line of reflection, on the opposite side, perpendicularly. (3) Moving the boat further moves the reflection the same amount — perpendicular distance is always equal. (4) Yes — given the line of reflection in the form y = mx + c, every point of the image can be	Stand at the DQB. Say: 'We are going to close this board today — officially. Every question we posted in Lesson 1 deserves a full answer. I will read each card, and we will answer it together using the mathematical language of this unit.' Read the first card. After reading, say: 'Who can answer this in one sentence — using the words line of reflection, perpendicular distance, or congruence?' [Allow 12 seconds WAIT TIME.] Cold-call a student. After the response, prompt the class: 'Thumbs up if you agree this question is fully answered. Thumbs sideways if partly. Thumbs down if you are still unsure.' If more than one-third show thumbs sideways or down, say: 'Let me add a final piece.' [Provide the resolution sentence from the five-question list.] After all questions: 'Write in your notebook: the single most important idea that explained the Lake Victoria phenomenon was _____. One sentence. 90 seconds.' [Allow 90 seconds.] Cold-call 3 students. Write their key idea sentences on the board.	DQB Completion Protocol — Formally closing each question on the DQB externalises the completion of the learning arc. Students experience the satisfaction of resolution, which reinforces the value of sustained inquiry. The thumbs vote makes agreement/disagreement visible and gives the teacher real-time data on residual misconceptions.	The thumbs vote is the primary formative indicator. Any question receiving more than one-third thumbs-sideways or thumbs-down responses is a signal that a concept has not been consolidated and requires teacher resolution before the exit task. The 'single most important idea' sentence is collected (or photographed) and reviewed: students who write a complete, mathematically precise sentence (e.g., 'Reflection maps every point to an equidistant position on the other side of the line of reflection, always producing a congruent image') are at the target level.

	predicted exactly. (5) For two shapes to be mathematically identical, they must be congruent — same shape, same size — which reflection always guarantees.			
Model Building Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cofunction Identities and Reflection (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — MODEL COMPARISON (10 minutes): Each student retrieves the boat-reflection model they drew in Lesson 1 (initial intuitive model) and places it alongside their most recent cumulative model revision. Using a structured comparison frame written on the board, they annotate both models: On the Lesson 1 model, they write in red: 'What I did not yet know.' On the final model, they write in green: 'What I now know and can prove.' Students then write a two-sentence Model Comparison Statement: 'In Lesson 1, I thought… Now I know… because…' Three students share their comparison statements with the class. PART B — EXIT EXPLANATION TASK (15 minutes, individual, written, in silence): Students write a full explanation of the anchoring phenomenon. The explanation must: (1) describe what happens to the boat's image at Lake Victoria; (2) explain WHY the image is always the same size, using the word 'isometry' and 'congruence'; (3) state the mathematical rule governing the position of every reflected point, referencing y = mx + c; (4) explain what it means for two shapes to be congruent; (5) connect to at least one of the three real-world application tasks from Phase 3. A sentence starter scaffold is available for students who request	Distribute Lesson 1 model sheets. Say: 'Place your Lesson 1 model and your final model side by side. In red, annotate your Lesson 1 model with what you did not yet know. In green, annotate your final model with what you now know and can prove. You have 5 minutes.' [Allow 5 minutes.] Then: 'Now write your two-sentence Model Comparison Statement in your notebook.' [Allow 3 minutes.] Cold-call 3 students to share. After each, ask the class: 'What mathematical term did they use that shows growth from Lesson 1?' [Allow 8 seconds WAIT TIME.] Transition to exit task: 'You now have 15 minutes to write your final explanation of the Lake Victoria phenomenon. This is individual and silent. Use every mathematical term we have built this unit. If you want the sentence starter, raise your hand quietly.' [Circulate during writing. Whisper prompts to struggling students: 'Can you mention the perpendicular distance rule?' or 'How does congruence appear in your explanation?'] At 12 minutes, give a 3-minute warning. Collect all exit explanations. Close with: 'You have just done what mathematicians do — you observed a natural phenomenon, asked questions, gathered evidence, and built a complete mathematical explanation. That is the power of mathematics.'	Model Comparison and Final Explanation Synthesis — Placing the Lesson 1 model beside the final model makes conceptual growth concrete and visible. The structured annotation (red/green) focuses attention on the specific ideas that changed. The exit explanation task requires students to integrate procedural knowledge (how to reflect a point), conceptual knowledge (why the image is congruent), and contextual knowledge (the Lake Victoria phenomenon) into a single coherent written account — the highest-order sensemaking demand of the unit.	The Exit Explanation Task is the summative formative assessment for the unit. Score against four criteria: (1) Phenomenon Description — does the student describe the Lake Victoria image accurately? (2) Mathematical Mechanism — do they correctly explain the perpendicular-distance rule and reference y = mx + c? (3) Congruence — do they define congruence and connect it to reflection as an isometry? (4) Real-World Connection — do they connect to at least one of the three application tasks? Students who address all four criteria with precise mathematical language are at the mastery level. Students who address two or three criteria with some imprecision are at the developing level and are flagged for follow-up review.

	support: 'When a fisherman's boat rests on the still water at Lake Victoria, the mirror image appears because…'			

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) FINAL EXPLANATION QUALITY — In the Exit Explanation Tasks, how many students addressed all four criteria (phenomenon description, mathematical mechanism, congruence definition, real-world connection)? What was the most common gap — procedural, conceptual, or contextual? (2) DQB COMPLETION — Were there any questions on the DQB that still received significant 'thumbs sideways/down' responses even after your resolution? If so, which concept was not fully consolidated across the eight lessons, and how would you restructure its placement in the storyline? (3) MODEL COMPARISON — When students annotated their Lesson 1 models in red, which misconceptions appeared most frequently? This data should inform how you introduce the reflection phenomenon in the next iteration of this unit. (4) APPLICATION TASKS — Which of the three tasks (Maasai bead pattern, gorge distance, building blueprint) generated the richest mathematical discussion? Which task caused the most confusion, and was the confusion procedural (e.g., computing perpendicular distance to y = x) or conceptual (e.g., understanding why congruence is guaranteed)? (5) STORYLINE COHERENCE — Did students in the Model Comparison phase show genuine conceptual growth from their Lesson 1 intuitive models, or did some students' final models still reflect primarily intuitive rather than mathematical reasoning? If the latter, at which lesson in the sequence did the conceptual shift fail to occur? Use these reflections to revise at least two lesson phases in the next delivery of this unit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	At dawn on the shore of Lake Victoria near Kisumu, a fisherman's boat casts a perfect mirror image on the still water. The reflected image appears identical in shape and size to the real boat — yet it is flipped. No matter how complex the boat's shape, the reflection is always the same size and always appears at an equal distance on the other side of the waterline. We also observed: a Maasai beadwork necklace where corresponding beads are equidistant from a central line of symmetry; a surveyor at the Rift Valley using congruent triangles to measure an inaccessible gorge width; and an architect's building blueprint where the two wings are mirror images of each other.
What did I learn?	We learned that: (1) Reflection maps every point in a figure to a corresponding point on the opposite side of the line of reflection, at exactly the same perpendicular distance — this is the rule the water surface follows. (2) The line of reflection can always be expressed in the form y = mx + c. (3) The image produced by reflection is always congruent to the original figure — same shape, same size — because reflection is an isometry (a distance-preserving transformation). (4) Congruent triangles can be used to calculate distances that cannot be measured directly. (5) These properties of reflection and congruence appear in Maasai design, land surveying, architecture, and the natural world.
How does this explain the phenomenon?	The perfect mirror image of the fisherman's boat at Lake Victoria is always the same size as the real boat because reflection is an isometry — it preserves all distances. The waterline acts as the line of reflection (expressible as y = mx + c). Every point on the boat maps to a corresponding image point that is exactly the same perpendicular distance from the waterline, but on the opposite side. Because every distance is preserved, the image is congruent to the original boat — identical in shape and size, but flipped. This is why the reflection never stretches, shrinks, or distorts the boat's image, no matter how far the boat is from the water's surface.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.